# BlindSpy

Classifying encrypted traffic (ISCXVPN2016) from 23 flow timing features

Report at milestone v0.2-tuned

Built 2026-10-29 from the report drafts at commit e124d38

# 1 Introduction

## 1.1 Problem

Most network traffic is now encrypted. Deep packet inspection, which classifies traffic by reading packet contents, no longer works when the contents are hidden. Network operators still need to know what kind of traffic a connection carries (for quality of service, access control and security monitoring) and whether it is routed through a VPN.

Applications leave a timing pattern even when their contents are encrypted. A voice call sends small packets at a steady rhythm, video streaming arrives in bursts, and a file transfer runs at full speed with almost no idle time. BlindSpy trains classifiers on these patterns only: 23 flow timing statistics per connection, never the payload.

We use the ISCXVPN2016 dataset [2] published with Draper-Gil et al. [1]. Each row is one flow described by 23 time-related features and a label. The dataset is provided in four versions, built with flow timeouts of 15, 30, 60 and 120 seconds. The timeout is a setting of the flow builder (it decides when an inactive flow is closed); it is not the duration of each flow.

## 1.2 Tasks

We follow the three tasks of the original paper [1], each scored on its own:

| Task | Description | Scored by |
|---|---|---|
| A1 | One binary classifier: VPN vs non-VPN | Precision and recall per class, macro F1, ROC-AUC |
| A2 | Two separate 7-class classifiers, one trained on VPN flows and one on non-VPN flows | Each model scored separately; never merged into one score |
| B | One 14-class classifier over VPN and non-VPN traffic together | Macro F1 and per-class scores over all 14 labels |

The seven traffic types are browsing, email, chat, streaming, file transfer, VoIP and P2P; task B uses each type twice (VPN and non-VPN).

## 1.3 What BlindSpy can and cannot claim

- It predicts the traffic type from aggregate timing statistics and never reads packet contents.
- It does not prove that traffic is encrypted.
- The data was captured in 2016, in a lab, with specific applications and one VPN setup. Results may not hold for current applications, VPN providers or networks. The conclusion repeats this.

## 1.4 Related work

| Study | Data and setup | Models | Reported result |
|---|---|---|---|
| Draper-Gil et al. 2016 [1] | ISCXVPN2016 ARFF files, tasks A1, A2, B, each timeout separately; Weka, 10-fold cross-validation | C4.5, KNN | A1 at 15 s, C4.5: precision 0.89 (VPN) and 0.906 (non-VPN); KNN: 0.848 and 0.846. A2 at 15 s, C4.5: average precision 0.84 (VPN model), 0.89 (non-VPN model). B, best timeout: average precision 0.783 (C4.5), 0.711 (KNN) |
| Bagui et al. 2017 [12] | Same time-related features | Random Forest, gradient boosting and others | About 90% (cited from the public abstract only; the full paper was not available to us, so no further detail is claimed) |
| Jorgensen et al. 2023 [14] | ISCXVPN2016 captures | Uncertainty-aware traffic labelling | Documents quality problems in the dataset's captures (see Section 7) |
| Isnan et al. 2026 [13] | Kaggle copy of the dataset, 59,706 rows; 18,719 duplicates removed, leaving 40,987; 14 classes; 80/20 split | Random Forest, 300 trees | 88.01% accuracy |

Paper values are taken from `reports/paper_baseline.csv`, which records each number's page and whether it is stated in the text or read from a figure.

**Why published scores (80–99%) are not directly comparable.** Studies on this dataset differ in the split (k-fold vs a single hold-out), the class setup (binary, 7-class or 14-class), whether duplicates are removed, and

whether flows from different timeout versions are mixed. Each of these can move accuracy by several points, so we compare only under matched protocols.

**The Isnan et al. data.** The Kaggle file used by Isnan et al. is the eight A2 files (VPN and non-VPN at all four timeouts) stacked together: compared as a multiset, its 59,706 rows equal the rows of those files (5,151 + 5,631 + 8,965 + 9,793 + 6,917 + 7,734 + 8,580 + 6,935), with no row left over and the same labels. It has 40,987 unique rows, exactly their count. 7,093 distinct flows appear in two to four timeout files, while only 3,676 rows are duplicates within one file (E00). So most of their 18,719 "duplicates" are the same flows repeated across timeout versions, and a random 80/20 split of the stacked file mixes the four flow-building settings in both training and test data. E16 reproduces their 88.01% (0.8796 on the same split) and shows that with the timeouts kept apart the same method scores 0.82-0.87 (mean 0.8430). A size control in E16 attributes that drop to training-set size, not to the mixing itself: stacking the four files gives 2.3-4.3 times as many training rows, and at equal size the mixed setup scores no higher (0.8395). Their score is therefore not comparable with a single-timeout score. The file also uses the A2 feature set (`total_fiat` and `total_biat` in place of the paper's `std_fiat` and `std_biat`; Section 2).

## 1.5 Contributions

1. A reproduction of the paper's C4.5 and KNN results under its own 10-fold protocol, compared precision with precision and recall with recall. [E02, E03, paper benchmark]
2. A leakage-safe protocol (stratified 70/15/15 split, cross-validation inside the training set only, test set scored once) applied to all three tasks and all four timeouts.
3. Stronger models (Random Forest, XGBoost, LightGBM), tuned with the same search budget as the best baseline. [E05–E08]
4. A check of the dataset itself: whether the ARFF column names match the data, duplicates, odd values and the class mix across timeouts, recorded as versioned data with a step-by-step cleaning log. [E00]
5. Analyses of what the models learned and how far the results can be trusted: feature-group ablation, SHAP, learning curve, split-seed stability and calibration. [E09, E13–E15]
6. A full history: every run, data version and experiment, traceable to a commit.

# 2 Dataset

## 2.1 Source and format

We use ISCXVPN2016 [2], published with Draper-Gil et al. [1] by the Canadian Institute for Cybersecurity, University of New Brunswick. The traffic was captured in 2016 in a lab, with and without a VPN, for seven traffic types: browsing, chat, streaming, mail, VoIP, P2P and file transfer. The authors turned the captures into flows with ISCXFlowMeter and published the flows as 20 ARFF files (Weka's text format). Each row is one flow: 23 numeric time-related features (times in microseconds, rates per second) and a label. We use only these files. They hold no payload, addresses, ports or timestamps. The SHA-256 hash of every file is recorded in `data/arff_sha256.txt`.

Each scenario is published at four flow timeouts (15, 30, 60 and 120 s). The timeout is a setting of the flow builder, not a flow duration. Flows longer than the timeout exist (up to 646 s) because the timeout closes a flow only after it has been inactive for that long.

## 2.2 Tasks and files

| Task | Files | Labels |
|---|---|---|
| A1 | `Scenario A1-ARFF`, 4 files | VPN, Non-VPN |
| A2-VPN | `Scenario A2-ARFF`, 4 `-VPN` files | 7 VPN traffic types |
| A2-NonVPN | `Scenario A2-ARFF`, 4 `-NO-VPN` files | 7 traffic types |
| B | `Scenario B-ARFF`, 4 files | 14 classes (7 types x VPN / non-VPN) |
| (B-AllinOne) | `Scenario B-ARFF`, 4 `-AllinOne` files | 7 types, VPN and non-VPN merged; inventoried, not a task |

| file | task | timeout | header | rows | classes | core_task |
|---|---|---|---|---|---|---|
| Scenario A1-ARFF/TimeBasedFeatures-Dataset-15s-VPN.arff | A1 | 15 | T | 18,758 | 2 | 1 |
| Scenario A1-ARFF/TimeBasedFeatures-Dataset-30s-VPN.arff | A1 | 30 | T | 14,651 | 2 | 1 |
| Scenario A1-ARFF/TimeBasedFeatures-Dataset-60s-VPN.arff | A1 | 60 | S | 15,515 | 2 | 1 |
| Scenario A1-ARFF/TimeBasedFeatures-Dataset-120s-VPN.arff | A1 | 120 | T | 10,782 | 2 | 1 |
| Scenario A2-ARFF/TimeBasedFeatures-Dataset-15s-VPN.arff | A2-VPN | 15 | T | 9,793 | 7 | 1 |
| Scenario A2-ARFF/TimeBasedFeatures-Dataset-30s-VPN.arff | A2-VPN | 30 | T | 7,734 | 7 | 1 |
| Scenario A2-ARFF/TimeBasedFeatures-Dataset-60s-VPN.arff | A2-VPN | 60 | T | 6,935 | 7 | 1 |
| Scenario A2-ARFF/TimeBasedFeatures-Dataset-120s-VPN.arff | A2-VPN | 120 | T | 5,631 | 7 | 1 |
| Scenario A2-ARFF/TimeBasedFeatures-Dataset-15s-NO-VPN.arff | A2-NonVPN | 15 | T | 8,965 | 7 | 1 |
| Scenario A2-ARFF/TimeBasedFeatures-Dataset-30s-NO-VPN.arff | A2-NonVPN | 30 | T | 6,917 | 7 | 1 |
| Scenario A2-ARFF/TimeBasedFeatures-Dataset-60s-NO-VPN.arff | A2-NonVPN | 60 | T | 8,580 | 7 | 1 |
| Scenario A2-ARFF/TimeBasedFeatures-Dataset-120s-NO-VPN.arff | A2-NonVPN | 120 | T | 5,151 | 7 | 1 |
| Scenario B-ARFF/TimeBasedFeatures-Dataset-15s.arff | B | 15 | T | 18,758 | 14 | 1 |

| file | task | timeout | header | rows | classes | core_task |
|---|---|---|---|---|---|---|
| Scenario B-ARFF/TimeBasedFeatures-Dataset-30s.arff | B | 30 | T | 14,651 | 14 | 1 |
| Scenario B-ARFF/TimeBasedFeatures-Dataset-60s.arff | B | 60 | S | 15,515 | 14 | 1 |
| Scenario B-ARFF/TimeBasedFeatures-Dataset-120s.arff | B | 120 | T | 10,782 | 14 | 1 |
| Scenario B-ARFF/TimeBasedFeatures-Dataset-15s-AllinOne.arff | B-AllinOne | 15 | T | 18,758 | 7 | 0 |
| Scenario B-ARFF/TimeBasedFeatures-Dataset-30s-AllinOne.arff | B-AllinOne | 30 | T | 14,651 | 7 | 0 |
| Scenario B-ARFF/TimeBasedFeatures-Dataset-60s-AllinOne.arff | B-AllinOne | 60 | S | 15,515 | 7 | 0 |
| Scenario B-ARFF/TimeBasedFeatures-Dataset-120s-AllinOne.arff | B-AllinOne | 120 | T | 10,782 | 7 | 0 |

Two features of the raw files matter for everything below. A1, A2 and B hold **the same flows** at each timeout: every A1 row matches an A2 row on seven integer features at all four timeouts (100%), and the labels agree key by key (0 disagreements). B and A1 are identical cell for cell except for one corrupt column (Section 2.4). BROWSING and VPN-BROWSING hold exactly 2,500 flows in every raw A2 file, which suggests the dataset authors capped those classes.

## 2.3 Column-name check

**Problem.** The first row of A1 15 s has `duration` = 9,368,711 us (9.4 s) but `total_fiat` = 16 and `total_biat` = 4. A total forward gap of 16 us in a 9-second flow is impossible. The paper's Table 2 [1] and the authors' 2016 poster [15] describe the forward and backward inter-arrival features as min, mean, max and std, with no totals. Yet 17 of the 20 headers list `total_fiat` and `total_biat` and no std. The three 60 s files of A1 and B have a second header that matches the paper.

**Method.** We read every file under both name lists. On every row we tested identities that hold whatever the traffic: min <= mean <= max for each gap statistic, 0 <= std <= max - min, mean flow gap x (packets - 1) = duration, and, for totals, total / mean + 1 per direction = packet count. Means of 10^7 or more are stored to 3 significant figures in the files, so a tolerance of 0.5% is allowed. We also tested the reading that `total_fiat` is a forward packet count.

| check | A1 15 s (T) | A1 15 s (S) | A2-VPN 15 s (T) | A2-VPN 15 s (S) | A2-NonVPN 15 s (T) | A2-NonVPN 15 s (S) | B 15 s (T) | B 15 s (S) |
|---|---|---|---|---|---|---|---|---|
| fiat: min <= mean <= max | 0.0 | 100.0 | 100.0 | 18.8 | 100.0 | 5.24 | 0.0 | 100.0 |
| biat: min <= mean <= max | 0.01 | 100.0 | 100.0 | 14.07 | 100.0 | 4.06 | 0.01 | 100.0 |
| flowiat: min <= mean <= max | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| active: min <= mean <= max | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| idle: min <= mean <= max | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| mean_flowiat x (packets - 1) = duration (+-1%) | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| bytes per packet in 40-1500 | 97.72 | 97.72 | 99.26 | 99.26 | 96.06 | 96.06 | 97.72 | 97.72 |
| min_flowiat <= min(min_fiat, min_biat) | 100.0 | 100.0 | 100.0 | 80.98 | 100.0 | 83.1 | 100.0 | 100.0 |
| total_fiat <= duration | 100.0 | - | 100.0 | - | 100.0 | - | 100.0 | - |
| total/mean + 1, fwd + bwd = packets (+-2%) | 0.0 | - | 100.0 | - | 100.0 | - | 0.0 | - |
| total_fiat as packet count (+-10%) | 1.98 | - | 0.0 | - | 0.04 | - | 1.98 | - |
| 0 <= std_fiat <= max - min | - | 100.0 | - | 0.0 | - | 0.0 | - | 100.0 |
| 0 <= std_biat <= max - min | - | 100.0 | - | 0.0 | - | 0.0 | - | 100.0 |

The same pattern holds at every timeout (`column_name_check` in the manifest has all 16 files).

**Result.** The headers are wrong for 9 of the 12 A1/B files, and the fault is a shift, not a shuffle. These files hold the paper's features in the order min, max, mean, std for fiat and biat, but their header names them total, min, max, mean. Under the corrected names every identity passes on 100% of rows; under the header names, min <= mean <= max passes on at most 0.02%. A2 files are named correctly and hold a different pair of features: `total_fiat` and `total_biat` (the sum of the forward and backward gaps) instead of `std_fiat` and `std_biat`. The packet-count reading passes on at most 2% of rows in any file, so it is ruled out. `COLUMN_MAP` in `src/blindspy/data.py` renames the 9 shifted files. It maps `total_fiat->min_fiat`, `total_biat->min_biat`, `min_fiat->max_fiat`, `min_biat->max_biat`, `max_fiat->mean_fiat`, `max_biat->mean_biat`, `mean_fiat->std_fiat`, `mean_biat->std_biat`; the other 15 names are unchanged. All feature names in this report (importances, SHAP, ablation) are the corrected ones.

**Units and value ranges.** Table 2 gives no units. The data does: with packets = flow packets per second x duration / 10^6, the mean flow gap x (packets - 1) equals the duration within 1% on 100% of rows in all 16 files, which holds only if times are in microseconds and rates per second. Every measured gap, active or idle value is at most the flow's duration (100% of values), as Table 2's definitions require. Under the corrected names the A1/B feature list is exactly Table 2's 23 features; the A2 list differs only in `total_fiat` and `total_biat` (`reports/data_checks_v1.json`).

| feature | min | median | p99 | max | exactly_-1 | other_negative | % <= duration |
|---|---|---|---|---|---|---|---|
| duration | 0 | 14,422,310 | 119,998,194 | 646,565,983 | 0 | 0 | - |
| min_fiat | -1 | 112 | 45,211,908 | 119,999,708 | 6,308 | 0 | 100.0 |
| min_biat | -1 | 45 | 45,213,098 | 119,999,620 | 10,522 | 0 | 100.0 |
| max_fiat | -1 | 1,650,890 | 104,244,407 | 646,287,731 | 6,308 | 0 | 100.0 |
| max_biat | -1 | 2,294,484 | 63,987,999 | 600,109,667 | 10,522 | 0 | 100.0 |
| mean_fiat | 0 | 158,066 | 45,200,000 | 152,000,000 | 0 | 0 | 100.0 |
| mean_biat | 0 | 83,359 | 45,200,000 | 120,000,000 | 0 | 0 | 100.0 |
| std_fiat | 0 | 27,808 | 34,000,000 | 264,000,000 | 0 | 0 | - |
| std_biat | 0 | 24,804 | 25,700,000 | 98,000,000 | 0 | 0 | - |
| flowPktsPerSecond | 0 | 8.96615 | 57,143 | 2,000,000 | 0 | 0 | - |
| flowBytesPerSecond | 0 | 1,695 | 6,140,974 | 617,000,000 | 0 | 0 | - |
| min_flowiat | -1,247 | 21 | 1,076,195 | 59,947,704 | 224 | 405 | 100.0 |
| max_flowiat | -1 | 962,728 | 109,547,639 | 646,287,731 | 45 | 0 | 100.0 |
| mean_flowiat | 0 | 144,230 | 19,900,000 | 64,700,000 | 0 | 0 | 100.0 |
| std_flowiat | 0 | 192,278 | 33,700,000 | 204,000,000 | 0 | 0 | - |
| min_active | -1 | 5,949,131 | 120,988,572 | 646,565,983 | 28,663 | 0 | 100.0 |
| mean_active | 0 | 0 | 110,000,000 | 647,000,000 | 0 | 0 | 100.0 |
| max_active | -1 | 10,316,202 | 122,147,146 | 646,565,983 | 28,663 | 0 | 100.0 |
| std_active | 0 | 0 | 23,200,000 | 184,000,000 | 0 | 0 | - |
| min_idle | -1 | 4,993,314 | 120,661,941 | 646,287,731 | 28,663 | 0 | 100.0 |
| mean_idle | 0 | 0 | 109,000,000 | 646,000,000 | 0 | 0 | 100.0 |
| max_idle | -1 | 10,000,610 | 121,615,880 | 646,287,731 | 28,663 | 0 | 100.0 |
| std_idle | 0 | 0 | 23,100,000 | 185,000,000 | 0 | 0 | - |

This also explains the earlier puzzle that A1 and A2 rows never match. As published (names ignored), 0 A1 rows equal an A2 row in all 23 values at any timeout, and only 2,364 of 18,758 at 15 s (1,501, 2,190 and 494 at 30, 60 and 120 s) contain the same set of values in another order. Two of the 23 values mean different things in the two families, and 9 of the 12 A1/B headers shift the names.

## 2.4 File-family decision

Because A1/B and A2 hold the same flows with two different feature pairs, we had three options: use each task's own files, rebuild A1 and B from the A2 files, or merge both into 25 features. We use **each task's own files**: A1 and B with the paper's 23 features, A2 with its 23. These are the files the paper trained on for each scenario, so our comparison with it stays matched. A rebuild would have replaced two of the paper's features, and a merge is ambiguous wherever a key repeats. For the two-step classifier in E10, both steps use the A1/B layout: the 7-class step is trained on B rows split by VPN (the same flows as A2).

One column needed repair. In A1 120 s, `flowPktsPerSecond` disagrees with A2 on 99.97% of matched rows, while B 120 s, identical to A1 120 s in the other 22 columns of every row, agrees with A2 on 100%. We restore the column from B 120 s, which changes 10,288 of 10,782 rows.

## 2.5 Cleaning rules

Every rule is in `configs/base.yaml` and is applied by `scripts/prepare_data.py` in this order. Each file gets one line per rule in the cleaning log, including rules that change nothing.

| # | Step | Rule | Reason |
|---|---|---|---|
| 1 | parse | Strip trailing commas from header lines; skip comma-only data lines | Export glitch; a comma-only line (one in each 60 s file of A1 and B) is not a flow |
| 2 | column map | Rename with `COLUMN_MAP` | Section 2.3 |
| 3 | restore | A1 120 s `flowPktsPerSecond` from B 120 s | Section 2.4 |
| 4-5 | missing, infinite | Count; none found | No imputation needed |
| 6-9 | -1 codes | Keep | -1 means "not measurable": fewer than 2 packets in a direction, a single-packet flow, or no active/idle period. The matching mean or total is 0 on the same rows |
| 10 | single-packet flows | Keep `duration` = 0 flows | Real flows; dropping them would bias the data toward long flows |
| 11 | negative `min_flowiat` | Keep (down to -1,247 us) | Out-of-order capture timestamps |
| 12 | rounding | Keep means, stds and rates >= 10^7 as stored | Already rounded to 3 significant figures in the source; can't be undone |
| 13 | packet size | Keep flows outside 40-1,500 B per packet | Header-only packets (down to 30 B) and segments merged at capture (up to about 7 kB) are real |
| 14 | exact duplicates | Drop within each file before splitting, keep the first copy | A copy in both train and test would score the model on rows it trained on |
| 15 | conflicting duplicates | Keep (one row per label) | Real ambiguity; removing it would make the test set easier than reality |
| 16-19 | not done | No normalization, no feature selection, no outlier removal, no class balancing | Scaling and resampling run inside the model pipeline on training folds only; all 23 features are kept |

Rows each kept-as-is rule applies to:

| rule | A1 | A2-VPN | A2-NonVPN | B |
|---|---|---|---|---|
| fiat_not_measurable | 8,846 | 5,120 | 3,726 | 8,846 |
| biat_not_measurable | 13,857 | 8,056 | 5,801 | 13,857 |
| flowiat_not_measurable | 2,108 | 1,205 | 903 | 2,108 |
| active_idle_not_measurable | 32,228 | 15,753 | 16,475 | 32,228 |
| single_packet | 2,110 | 1,207 | 903 | 2,110 |
| negative_flowiat | 586 | 300 | 286 | 586 |
| rounding | 15,217 | 7,979 | 7,234 | 15,217 |
| packet_size | 1,380 | 397 | 983 | 1,380 |

## 2.6 Dataset summary

| file | initial_rows | comma_only_lines_skipped | exact_duplicates_removed | conflicting_rows_kept | conflicting_groups | final_rows | missing_values | features | classes | largest_to_smallest_class | train | val | test |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A1 15 s | 18,758 | 0 | 720 | 46 | 23 | 18,038 | 0 | 23 | 2 | 1.08:1 | 12,626 | 2,706 | 2,706 |
| A1 30 s | 14,651 | 0 | 1,041 | 44 | 22 | 13,610 | 0 | 23 | 2 | 1.03:1 | 9,526 | 2,042 | 2,042 |
| A1 60 s | 15,516 | 1 | 838 | 94 | 47 | 14,677 | 0 | 23 | 2 | 1.26:1 | 10,273 | 2,202 | 2,202 |
| A1 120 s | 10,782 | 0 | 1,258 | 62 | 31 | 9,524 | 0 | 23 | 2 | 1.08:1 | 6,666 | 1,429 | 1,429 |
| A2-VPN 15 s | 9,793 | 0 | 413 | 53 | 24 | 9,380 | 0 | 23 | 7 | 5.79:1 | 6,566 | 1,407 | 1,407 |
| A2-VPN 30 s | 7,734 | 0 | 790 | 39 | 17 | 6,944 | 0 | 23 | 7 | 8.39:1 | 4,860 | 1,042 | 1,042 |
| A2-VPN 60 s | 6,935 | 0 | 412 | 61 | 28 | 6,523 | 0 | 23 | 7 | 12.50:1 | 4,565 | 979 | 979 |
| A2-VPN 120 s | 5,631 | 0 | 983 | 111 | 52 | 4,648 | 0 | 23 | 7 | 15.48:1 | 3,252 | 698 | 698 |
| A2-NonVPN 15 s | 8,965 | 0 | 271 | 10 | 3 | 8,694 | 0 | 23 | 7 | 11.85:1 | 6,085 | 1,304 | 1,305 |
| A2-NonVPN 30 s | 6,917 | 0 | 223 | 9 | 3 | 6,694 | 0 | 23 | 7 | 32.47:1 | 4,685 | 1,004 | 1,005 |
| A2-NonVPN 60 s | 8,580 | 0 | 374 | 34 | 15 | 8,206 | 0 | 23 | 7 | 11.31:1 | 5,744 | 1,231 | 1,231 |
| A2-NonVPN 120 s | 5,151 | 0 | 210 | 9 | 3 | 4,941 | 0 | 23 | 7 | 14.37:1 | 3,458 | 741 | 742 |
| B 15 s | 18,758 | 0 | 684 | 100 | 41 | 18,074 | 0 | 23 | 14 | 11.85:1 | 12,651 | 2,711 | 2,712 |
| B 30 s | 14,651 | 0 | 1,013 | 83 | 33 | 13,638 | 0 | 23 | 14 | 32.47:1 | 9,546 | 2,046 | 2,046 |
| B 60 s | 15,516 | 1 | 786 | 174 | 75 | 14,729 | 0 | 23 | 14 | 12.69:1 | 10,309 | 2,210 | 2,210 |
| B 120 s | 10,782 | 0 | 1,193 | 173 | 77 | 9,589 | 0 | 23 | 14 | 17.24:1 | 6,711 | 1,439 | 1,439 |

## 2.7 Data funnel (data v1)

Only two steps remove rows: the parser (two comma-only lines in total) and exact de-duplication (11,209 rows). Over the 16 core files, 179,120 raw data lines become 167,909 cleaned flows; no value is changed except the restored A1 120 s column.

| file | raw data lines | parsed | after duplicates | final | train | val | test |
|---|---|---|---|---|---|---|---|
| A1 15 s | 18,758 | 18,758 | 18,038 | 18,038 | 12,626 | 2,706 | 2,706 |
| A1 30 s | 14,651 | 14,651 | 13,610 | 13,610 | 9,526 | 2,042 | 2,042 |
| A1 60 s | 15,516 | 15,515 | 14,677 | 14,677 | 10,273 | 2,202 | 2,202 |
| A1 120 s | 10,782 | 10,782 | 9,524 | 9,524 | 6,666 | 1,429 | 1,429 |
| A2-VPN 15 s | 9,793 | 9,793 | 9,380 | 9,380 | 6,566 | 1,407 | 1,407 |
| A2-VPN 30 s | 7,734 | 7,734 | 6,944 | 6,944 | 4,860 | 1,042 | 1,042 |
| A2-VPN 60 s | 6,935 | 6,935 | 6,523 | 6,523 | 4,565 | 979 | 979 |
| A2-VPN 120 s | 5,631 | 5,631 | 4,648 | 4,648 | 3,252 | 698 | 698 |
| A2-NonVPN 15 s | 8,965 | 8,965 | 8,694 | 8,694 | 6,085 | 1,304 | 1,305 |
| A2-NonVPN 30 s | 6,917 | 6,917 | 6,694 | 6,694 | 4,685 | 1,004 | 1,005 |
| A2-NonVPN 60 s | 8,580 | 8,580 | 8,206 | 8,206 | 5,744 | 1,231 | 1,231 |
| A2-NonVPN 120 s | 5,151 | 5,151 | 4,941 | 4,941 | 3,458 | 741 | 742 |
| B 15 s | 18,758 | 18,758 | 18,074 | 18,074 | 12,651 | 2,711 | 2,712 |
| B 30 s | 14,651 | 14,651 | 13,638 | 13,638 | 9,546 | 2,046 | 2,046 |
| B 60 s | 15,516 | 15,515 | 14,729 | 14,729 | 10,309 | 2,210 | 2,210 |

| file | raw data lines | parsed | after duplicates | final | train | val | test |
|---|---|---|---|---|---|---|---|
| B 120 s | 10,782 | 10,782 | 9,589 | 9,589 | 6,711 | 1,439 | 1,439 |
| **All 16 files** | 179,120 | 179,118 | 167,909 | 167,909 | 117,523 | 25,191 | 25,195 |

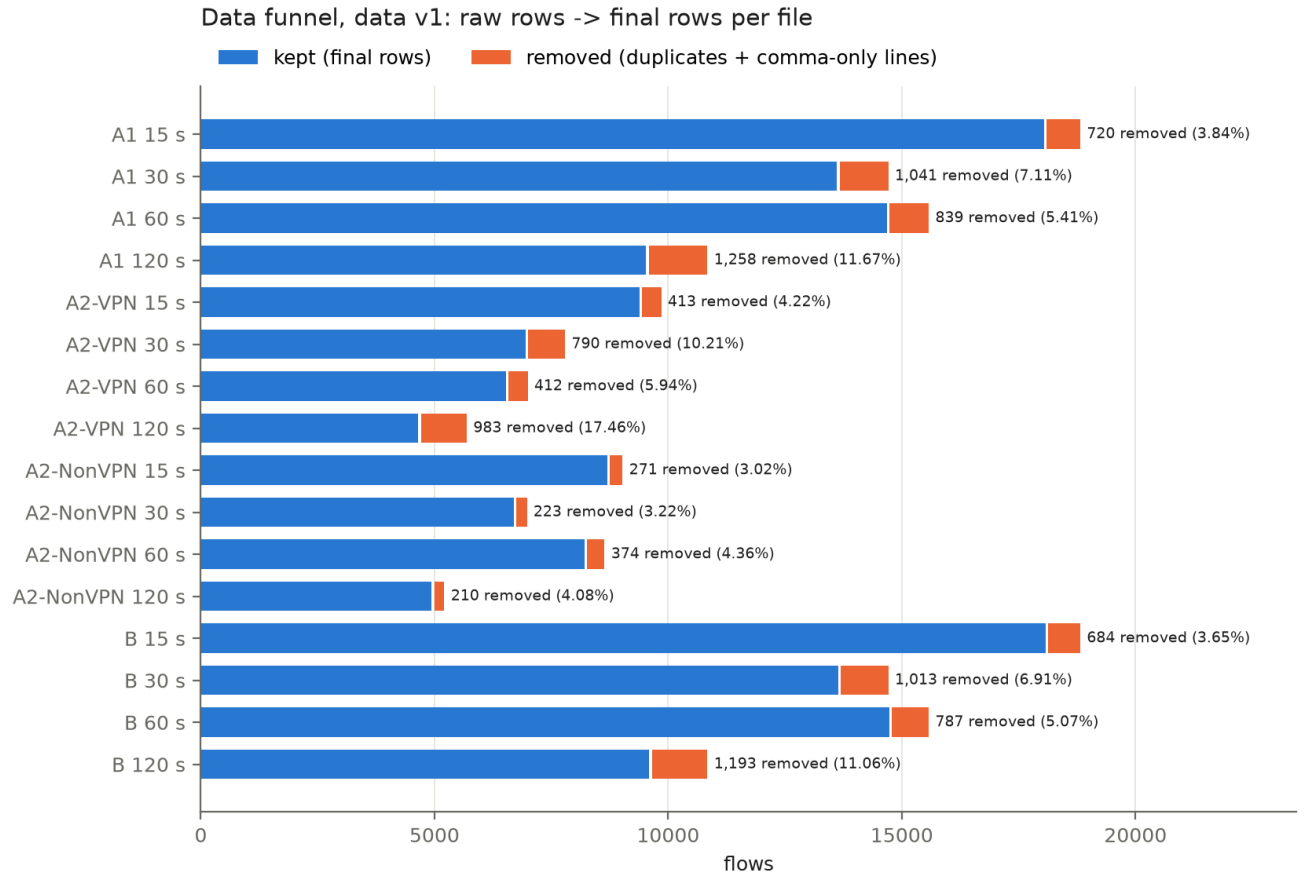



*Figure 1. Data funnel*

Removals are uneven by class, so they change the class balance.

A1

| class | 15 s | 30 s | 60 s | 120 s |
|---|---|---|---|---|
| Non-VPN | 278 of 8,965 | 229 of 6,917 | 393 of 8,580 | 216 of 5,151 |
| VPN | 442 of 9,793 | 812 of 7,734 | 445 of 6,935 | 1,042 of 5,631 |

A2-VPN

| class | 15 s | 30 s | 60 s | 120 s |
|---|---|---|---|---|
| VPN-BROWSING | 0 of 2,500 | 0 of 2,500 | 37 of 2,500 | 256 of 2,500 |
| VPN-CHAT | 38 of 1,196 | 38 of 780 | 36 of 514 | 39 of 349 |
| VPN-STREAMING | 0 of 475 | 0 of 298 | 0 of 197 | 0 of 145 |
| VPN-MAIL | 59 of 491 | 483 of 848 | 59 of 383 | 420 of 722 |
| VPN-VOIP | 5 of 2,271 | 5 of 1,299 | 9 of 1,620 | 5 of 386 |
| VPN-P2P | 262 of 928 | 230 of 851 | 218 of 823 | 215 of 813 |
| VPN-FT | 49 of 1,932 | 34 of 1,158 | 53 of 898 | 48 of 716 |

A2-NonVPN

| class | 15 s | 30 s | 60 s | 120 s |
|---|---|---|---|---|
| BROWSING | 0 of 2,500 | 0 of 2,500 | 0 of 2,500 | 0 of 2,500 |
| CHAT | 3 of 890 | 7 of 595 | 24 of 778 | 3 of 242 |
| STREAMING | 32 of 482 | 31 of 342 | 31 of 252 | 31 of 208 |
| MAIL | 11 of 249 | 4 of 81 | 117 of 849 | 11 of 185 |
| VOIP | 5 of 2,826 | 3 of 1,438 | 36 of 1,829 | 3 of 392 |
| P2P | 59 of 1,000 | 7 of 1,000 | 4 of 1,000 | 2 of 1,000 |
| FT | 161 of 1,018 | 171 of 961 | 162 of 1,372 | 160 of 624 |

B

| class | 15 s | 30 s | 60 s | 120 s |
|---|---|---|---|---|
| BROWSING | 0 of 2,500 | 0 of 2,500 | 0 of 2,500 | 0 of 2,500 |
| CHAT | 3 of 890 | 7 of 595 | 24 of 778 | 3 of 242 |
| STREAMING | 32 of 482 | 31 of 342 | 31 of 252 | 31 of 208 |
| MAIL | 11 of 249 | 4 of 81 | 117 of 849 | 11 of 185 |
| VOIP | 5 of 2,826 | 3 of 1,438 | 36 of 1,829 | 3 of 392 |
| P2P | 59 of 1,000 | 7 of 1,000 | 4 of 1,000 | 2 of 1,000 |
| FT | 161 of 1,018 | 171 of 961 | 162 of 1,372 | 160 of 624 |
| VPN-BROWSING | 0 of 2,500 | 0 of 2,500 | 37 of 2,500 | 256 of 2,500 |
| VPN-CHAT | 38 of 1,196 | 38 of 780 | 36 of 514 | 39 of 349 |
| VPN-STREAMING | 0 of 475 | 0 of 298 | 0 of 197 | 0 of 145 |
| VPN-MAIL | 59 of 491 | 483 of 848 | 59 of 383 | 420 of 722 |
| VPN-VOIP | 5 of 2,271 | 5 of 1,299 | 9 of 1,620 | 5 of 386 |
| VPN-P2P | 262 of 928 | 230 of 851 | 218 of 823 | 215 of 813 |
| VPN-FT | 49 of 1,932 | 34 of 1,158 | 53 of 898 | 48 of 716 |

| file | groups | rows | test rows with a conflicting twin in train |
|---|---|---|---|
| A1 15 s | 23 | 46 | 4 |
| A1 30 s | 22 | 44 | 8 |
| A1 60 s | 47 | 94 | 9 |
| A1 120 s | 31 | 62 | 7 |
| A2-VPN 15 s | 24 | 53 | 9 |
| A2-VPN 30 s | 17 | 39 | 4 |
| A2-VPN 60 s | 28 | 61 | 6 |
| A2-VPN 120 s | 52 | 111 | 11 |
| A2-NonVPN 15 s | 3 | 10 | 3 |
| A2-NonVPN 30 s | 3 | 9 | 2 |
| A2-NonVPN 60 s | 15 | 34 | 6 |
| A2-NonVPN 120 s | 3 | 9 | 1 |
| B 15 s | 41 | 100 | 15 |
| B 30 s | 33 | 83 | 15 |
| B 60 s | 75 | 174 | 25 |
| B 120 s | 77 | 173 | 21 |

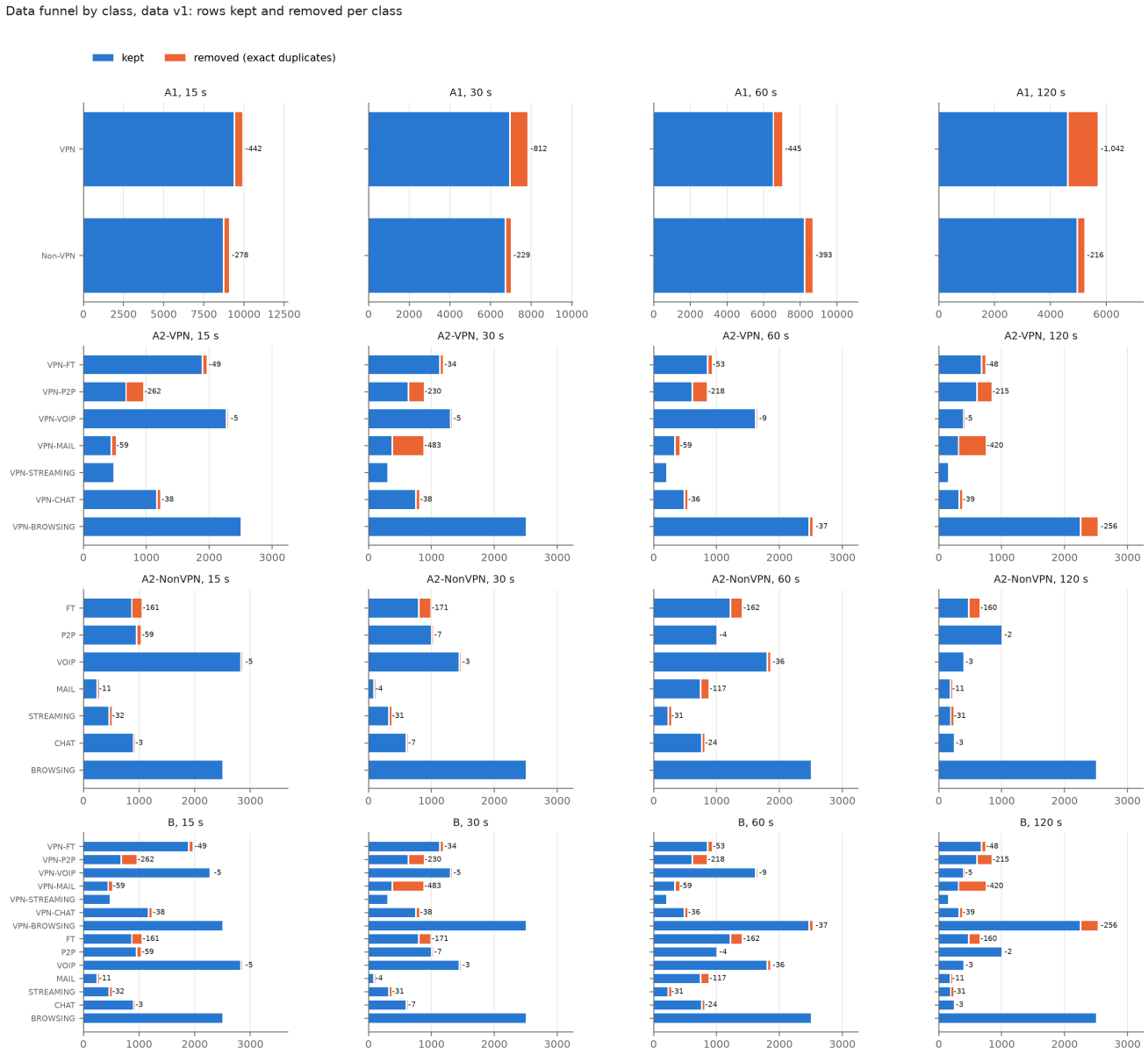



*Figure 2. Data funnel by class*

Per class, from raw rows to the final train / validation / test sets:

A1

| class | timeout | raw | final | train | val | test |
|---|---|---|---|---|---|---|
| Non-VPN | 15 s | 8,965 | 8,687 | 6,081 | 1,303 | 1,303 |
| VPN | 15 s | 9,793 | 9,351 | 6,545 | 1,403 | 1,403 |
| Non-VPN | 30 s | 6,917 | 6,688 | 4,681 | 1,004 | 1,003 |
| VPN | 30 s | 7,734 | 6,922 | 4,845 | 1,038 | 1,039 |
| Non-VPN | 60 s | 8,580 | 8,187 | 5,731 | 1,228 | 1,228 |
| VPN | 60 s | 6,935 | 6,490 | 4,542 | 974 | 974 |
| Non-VPN | 120 s | 5,151 | 4,935 | 3,454 | 741 | 740 |
| VPN | 120 s | 5,631 | 4,589 | 3,212 | 688 | 689 |

A2-VPN

| class | timeout | raw | final | train | val | test |
|---|---|---|---|---|---|---|
| VPN-BROWSING | 15 s | 2,500 | 2,500 | 1,750 | 375 | 375 |

| class | timeout | raw | final | train | val | test |
|---|---|---|---|---|---|---|
| VPN-CHAT | 15 s | 1,196 | 1,158 | 810 | 174 | 174 |
| VPN-STREAMING | 15 s | 475 | 475 | 333 | 71 | 71 |
| VPN-MAIL | 15 s | 491 | 432 | 302 | 65 | 65 |
| VPN-VOIP | 15 s | 2,271 | 2,266 | 1,586 | 340 | 340 |
| VPN-P2P | 15 s | 928 | 666 | 466 | 100 | 100 |
| VPN-FT | 15 s | 1,932 | 1,883 | 1,319 | 282 | 282 |
| VPN-BROWSING | 30 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| VPN-CHAT | 30 s | 780 | 742 | 520 | 111 | 111 |
| VPN-STREAMING | 30 s | 298 | 298 | 208 | 45 | 45 |
| VPN-MAIL | 30 s | 848 | 365 | 255 | 55 | 55 |
| VPN-VOIP | 30 s | 1,299 | 1,294 | 906 | 194 | 194 |
| VPN-P2P | 30 s | 851 | 621 | 435 | 93 | 93 |
| VPN-FT | 30 s | 1,158 | 1,124 | 786 | 169 | 169 |
| VPN-BROWSING | 60 s | 2,500 | 2,463 | 1,724 | 369 | 370 |
| VPN-CHAT | 60 s | 514 | 478 | 334 | 72 | 72 |
| VPN-STREAMING | 60 s | 197 | 197 | 139 | 29 | 29 |
| VPN-MAIL | 60 s | 383 | 324 | 227 | 49 | 48 |
| VPN-VOIP | 60 s | 1,620 | 1,611 | 1,127 | 242 | 242 |
| VPN-P2P | 60 s | 823 | 605 | 423 | 91 | 91 |
| VPN-FT | 60 s | 898 | 845 | 591 | 127 | 127 |
| VPN-BROWSING | 120 s | 2,500 | 2,244 | 1,570 | 337 | 337 |
| VPN-CHAT | 120 s | 349 | 310 | 216 | 47 | 47 |
| VPN-STREAMING | 120 s | 145 | 145 | 101 | 22 | 22 |
| VPN-MAIL | 120 s | 722 | 302 | 212 | 45 | 45 |
| VPN-VOIP | 120 s | 386 | 381 | 267 | 57 | 57 |
| VPN-P2P | 120 s | 813 | 598 | 418 | 90 | 90 |
| VPN-FT | 120 s | 716 | 668 | 468 | 100 | 100 |

A2-NonVPN

| class | timeout | raw | final | train | val | test |
|---|---|---|---|---|---|---|
| BROWSING | 15 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| CHAT | 15 s | 890 | 887 | 621 | 133 | 133 |
| STREAMING | 15 s | 482 | 450 | 315 | 67 | 68 |
| MAIL | 15 s | 249 | 238 | 166 | 36 | 36 |
| VOIP | 15 s | 2,826 | 2,821 | 1,975 | 423 | 423 |
| P2P | 15 s | 1,000 | 941 | 659 | 141 | 141 |
| FT | 15 s | 1,018 | 857 | 599 | 129 | 129 |
| BROWSING | 30 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| CHAT | 30 s | 595 | 588 | 412 | 88 | 88 |
| STREAMING | 30 s | 342 | 311 | 217 | 47 | 47 |
| MAIL | 30 s | 81 | 77 | 53 | 12 | 12 |

| class | timeout | raw | final | train | val | test |
|---|---|---|---|---|---|---|
| VOIP | 30 s | 1,438 | 1,435 | 1,005 | 215 | 215 |
| P2P | 30 s | 1,000 | 993 | 695 | 149 | 149 |
| FT | 30 s | 961 | 790 | 553 | 118 | 119 |
| BROWSING | 60 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| CHAT | 60 s | 778 | 754 | 528 | 113 | 113 |
| STREAMING | 60 s | 252 | 221 | 155 | 33 | 33 |
| MAIL | 60 s | 849 | 732 | 512 | 110 | 110 |
| VOIP | 60 s | 1,829 | 1,793 | 1,255 | 269 | 269 |
| P2P | 60 s | 1,000 | 996 | 697 | 150 | 149 |
| FT | 60 s | 1,372 | 1,210 | 847 | 181 | 182 |
| BROWSING | 120 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| CHAT | 120 s | 242 | 239 | 167 | 36 | 36 |
| STREAMING | 120 s | 208 | 177 | 124 | 26 | 27 |
| MAIL | 120 s | 185 | 174 | 122 | 26 | 26 |
| VOIP | 120 s | 392 | 389 | 273 | 58 | 58 |
| P2P | 120 s | 1,000 | 998 | 698 | 150 | 150 |
| FT | 120 s | 624 | 464 | 324 | 70 | 70 |

B

| class | timeout | raw | final | train | val | test |
|---|---|---|---|---|---|---|
| BROWSING | 15 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| CHAT | 15 s | 890 | 887 | 621 | 133 | 133 |
| STREAMING | 15 s | 482 | 450 | 315 | 68 | 67 |
| MAIL | 15 s | 249 | 238 | 166 | 36 | 36 |
| VOIP | 15 s | 2,826 | 2,821 | 1,975 | 423 | 423 |
| P2P | 15 s | 1,000 | 941 | 659 | 141 | 141 |
| FT | 15 s | 1,018 | 857 | 600 | 128 | 129 |
| VPN-BROWSING | 15 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| VPN-CHAT | 15 s | 1,196 | 1,158 | 810 | 174 | 174 |
| VPN-STREAMING | 15 s | 475 | 475 | 333 | 71 | 71 |
| VPN-MAIL | 15 s | 491 | 432 | 302 | 65 | 65 |
| VPN-VOIP | 15 s | 2,271 | 2,266 | 1,586 | 340 | 340 |
| VPN-P2P | 15 s | 928 | 666 | 466 | 100 | 100 |
| VPN-FT | 15 s | 1,932 | 1,883 | 1,318 | 282 | 283 |
| BROWSING | 30 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| CHAT | 30 s | 595 | 588 | 412 | 88 | 88 |
| STREAMING | 30 s | 342 | 311 | 217 | 47 | 47 |
| MAIL | 30 s | 81 | 77 | 54 | 11 | 12 |
| VOIP | 30 s | 1,438 | 1,435 | 1,005 | 215 | 215 |
| P2P | 30 s | 1,000 | 993 | 695 | 149 | 149 |
| FT | 30 s | 961 | 790 | 553 | 119 | 118 |

| class | timeout | raw | final | train | val | test |
|---|---|---|---|---|---|---|
| VPN-BROWSING | 30 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| VPN-CHAT | 30 s | 780 | 742 | 520 | 111 | 111 |
| VPN-STREAMING | 30 s | 298 | 298 | 208 | 45 | 45 |
| VPN-MAIL | 30 s | 848 | 365 | 255 | 55 | 55 |
| VPN-VOIP | 30 s | 1,299 | 1,294 | 906 | 194 | 194 |
| VPN-P2P | 30 s | 851 | 621 | 435 | 93 | 93 |
| VPN-FT | 30 s | 1,158 | 1,124 | 786 | 169 | 169 |
| BROWSING | 60 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| CHAT | 60 s | 778 | 754 | 528 | 113 | 113 |
| STREAMING | 60 s | 252 | 221 | 155 | 33 | 33 |
| MAIL | 60 s | 849 | 732 | 512 | 110 | 110 |
| VOIP | 60 s | 1,829 | 1,793 | 1,255 | 269 | 269 |
| P2P | 60 s | 1,000 | 996 | 698 | 149 | 149 |
| FT | 60 s | 1,372 | 1,210 | 847 | 182 | 181 |
| VPN-BROWSING | 60 s | 2,500 | 2,463 | 1,724 | 369 | 370 |
| VPN-CHAT | 60 s | 514 | 478 | 334 | 72 | 72 |
| VPN-STREAMING | 60 s | 197 | 197 | 138 | 30 | 29 |
| VPN-MAIL | 60 s | 383 | 324 | 227 | 48 | 49 |
| VPN-VOIP | 60 s | 1,620 | 1,611 | 1,127 | 242 | 242 |
| VPN-P2P | 60 s | 823 | 605 | 423 | 91 | 91 |
| VPN-FT | 60 s | 898 | 845 | 591 | 127 | 127 |
| BROWSING | 120 s | 2,500 | 2,500 | 1,750 | 375 | 375 |
| CHAT | 120 s | 242 | 239 | 167 | 36 | 36 |
| STREAMING | 120 s | 208 | 177 | 124 | 26 | 27 |
| MAIL | 120 s | 185 | 174 | 122 | 26 | 26 |
| VOIP | 120 s | 392 | 389 | 273 | 58 | 58 |
| P2P | 120 s | 1,000 | 998 | 698 | 150 | 150 |
| FT | 120 s | 624 | 464 | 324 | 70 | 70 |
| VPN-BROWSING | 120 s | 2,500 | 2,244 | 1,570 | 337 | 337 |
| VPN-CHAT | 120 s | 349 | 310 | 217 | 47 | 46 |
| VPN-STREAMING | 120 s | 145 | 145 | 101 | 22 | 22 |
| VPN-MAIL | 120 s | 722 | 302 | 212 | 45 | 45 |
| VPN-VOIP | 120 s | 386 | 381 | 267 | 57 | 57 |
| VPN-P2P | 120 s | 813 | 598 | 418 | 90 | 90 |
| VPN-FT | 120 s | 716 | 668 | 468 | 100 | 100 |

**In sentences.** For task A1 at 15 s, 18,758 flows were parsed. After removing 720 exact duplicates (442 VPN, 278 non-VPN), 18,038 unique flows remained. All 23 features were kept without feature selection or normalization, and the flows were split 12,626 / 2,706 / 2,706 into training, validation and test sets. Duplicates are most common at the 120 s timeout: 1,258 of 10,782 A1 flows (11.67%) and 983 of 5,631 A2-VPN flows (17.46%). They hit some classes much harder than others: VPN-MAIL lost 483 of 848 flows at 30 s and 420 of 722 at 120 s, FT lost 160-171 flows at every timeout, and BROWSING lost none. A1 loses slightly more rows than B at the same timeout (720 vs 684 at 15 s) because flows that share features but differ in traffic type

become exact duplicates once both are labelled with the same VPN / non-VPN label. After cleaning, no exact duplicate crosses train and test in any file; 1-25 test flows per file have a twin in training with the same features but a different label.

**Deliberately not done, and why.** No values were imputed (none are missing). No -1 code was replaced and nothing was set to zero, because -1 has a meaning. No flag columns were added, since the matching zero mean already marks the same rows and flags would change the paper's 23-feature set. No rows were dropped as outliers, because long, bursty or single-packet flows are real traffic. No features were scaled or selected and no classes were rebalanced at data level; all of that happens inside model pipelines, fitted on training data only (Section 3).

## 2.8 Timeouts

Within each task the feature names, their order and their units are the same at all four timeouts. The number of flows falls as the timeout grows (A1: 18,038 at 15 s to 9,524 at 120 s), and the class mix moves a lot. One class's share changes by up to 24.6 percentage points in A2-NonVPN (VOIP 32.5% at 15 s vs 7.9% at 120 s; BROWSING 28.8% vs 50.6%), 21.6 in A2-VPN, 12.2 in B and 7.6 in A1. A difference in scores between timeouts is therefore confounded by class mix, and we do not attribute it to the timeout alone.




*Figure 3. Class mix per timeout*

**The 60 s files differ in what some classes look like.** At 15, 30 and 120 s, non-VPN VOIP flows send about 100 packets per second with a 20 ms median forward gap, and VPN-VOIP flows about 2 with a gap of about 960 ms. At 60 s the two are almost reversed (2.1 packets/s and 962 ms for VOIP, 16.5 packets/s and 18 ms for VPN-VOIP), and non-VPN FT drops from 1,237-3,739 to 56.5 packets/s. It is not a label swap. Of 7,041 flows found identically in more than one timeout file, 210 change label between timeouts; they are mostly 2-packet flows, and no VoIP pair is among them. The 60 s files are also the only ones exported with the paper-style header, so they may come from a different export run. The ARFF files can't settle the cause. Results at 60 s are therefore read with care and the issue is listed under Limitations (Section 7).

| class, timeout | flows | median packets per second | median mean_fiat (us) | % mean_fiat 15-25 ms | % no active/idle period |
|---|---|---|---|---|---|
| VOIP 15s | 2,821 | 100.1 | 19994.9 | 78.1 | 99.4 |
| VOIP 30s | 1,435 | 100.0 | 19998.1 | 77.1 | 98.9 |
| VOIP 60s | 1,793 | 2.1 | 961624.6 | 31.2 | 51.5 |
| VOIP 120s | 389 | 100.0 | 19999.8 | 73.5 | 95.4 |
| FT 15s | 857 | 1236.7 | 1042.8 | 1.2 | 69.4 |
| FT 30s | 790 | 2155.9 | 768.0 | 1.5 | 72.0 |
| FT 60s | 1,210 | 56.5 | 34444.3 | 7.8 | 80.1 |

| class, timeout | flows | median packets per second | median mean_fiat (us) | % mean_fiat 15-25 ms | % no active/idle period |
|---|---|---|---|---|---|
| FT 120s | 464 | 3738.5 | 507.5 | 0.4 | 78.9 |
| VPN-VOIP 15s | 2,266 | 2.2 | 960175.4 | 42.9 | 64.4 |
| VPN-VOIP 30s | 1,294 | 2.1 | 960284.3 | 37.7 | 56.6 |
| VPN-VOIP 60s | 1,611 | 16.5 | 18295.3 | 15.7 | 71.6 |
| VPN-VOIP 120s | 381 | 2.1 | 960800.7 | 32.3 | 49.3 |
| VPN-FT 15s | 1,883 | 7.1 | 192693.4 | 0.7 | 55.1 |
| VPN-FT 30s | 1,124 | 4.0 | 416771.2 | 0.8 | 35.0 |
| VPN-FT 60s | 845 | 6.1 | 286898.0 | 1.4 | 45.1 |
| VPN-FT 120s | 668 | 7.5 | 187957.4 | 2.1 | 48.8 |

## 2.9 The Kaggle copy used by Isnan et al.

The Kaggle file used by Isnan et al. [13] has 59,706 rows and the A2 header. Compared as a multiset of rows, it equals the eight A2 files (VPN and non-VPN at all four timeouts) stacked: no row is left over at 12 significant digits, and the labels are the same. It has 40,987 unique rows, their count after removing 18,719 duplicates. Only 3,676 of those duplicates lie within one file; 7,093 distinct flows appear in two (2,475), three (1,286) or four (3,332) timeout files. Most of their "duplicates" are therefore the same flow built with different timeouts, and a random 80/20 split of the stacked file mixes all four timeout settings in training and test. The file also carries the A2 feature pair (`total_*`), not the paper's (`std_*`). E16 reproduces their setup and measures the effect of keeping timeouts apart (Section 4): the drop it finds is explained by training-set size, not by the mixing itself.

## 2.10 Data versions

| Version | Date | Author | What changed | Files | Rows (raw lines -> final) | Commit |
|---|---|---|---|---|---|---|
| v1 | 2026-10-27 | utsav-s-r | First cleaned version | 16 core files (A1, A2-VPN, A2-NonVPN, B x 15/30/60/120 s) | 179,120 -> 167,909 | 0a36896 |

Figures from the exploration notebook (`notebooks/01_eda.ipynb`): class counts per task and timeout (`class_counts_v1.png`), feature distributions by class at every timeout (`feature_boxes_<task>_<timeout>s.png`, 16 figures) and feature correlations (`correlation_<task>_15s.png`, all four tasks), the data funnel by class (`data_funnel_by_class_v1.png`). The notebook records two findings that matter later. Classes differ by orders of magnitude on rate and gap features (VPN-STREAMING: median 202 packets/s; VPN-VOIP: 2.2). Many features are near-duplicates: 19 of 253 pairs in A1 and B (27 in A2-VPN, 20 in A2-NonVPN) have Spearman |rho| >= 0.9, mostly within the active/idle group. Feature importances will therefore be shared out among correlated features.

# 3 Method

## 3.1 Pipeline overview

All code is in the project repository (Python 3.12). Dependencies are managed with uv; exact package versions are fixed in `uv.lock`, so every team member and every rerun uses identical library versions. The main libraries are pandas and NumPy (data), liac-arff and pyarrow (reading ARFF, writing Parquet), scikit-learn [3] (models, pipelines, metrics), MLflow [8] (run tracking), matplotlib and seaborn (figures) and SHAP [6] (explanations). XGBoost [4], LightGBM [5], Optuna [7] and imbalanced-learn [9] are added at the experiments that first need them.

The flow of data and results is:

1. **Raw data.** The 20 ARFF files of ISCXVPN2016 [2] are kept unchanged; their SHA-256 hashes are recorded in `data/arff_sha256.txt` and checked on every machine.
2. **Preparation.** `scripts/prepare_data.py` parses the files, applies the verified column-name map and the cleaning rules, and writes one Parquet file per task and timeout to a versioned folder `data/processed/vN/`. [Column-name check and cleaning rules: E00.]
3. **Training.** `scripts/run_experiment.py` reads one YAML configuration file, builds a scikit-learn Pipeline (preprocessing + model), runs 5-fold cross-validation on the training set, fits on the training set and scores the validation set.
4. **Tracking.** Every run is logged to MLflow (Section 3.6).
5. **Reporting.** `scripts/export_history.py` exports every run to `reports/history.csv`; `scripts/results_tables.py` builds every result table from it, and the notebooks (written for Phase 3 by `scripts/make_notebooks.py`) draw the figures from it, so no number in this report is copied by hand.

## 3.2 Data versions

Any change that alters the cleaned data (a cleaning rule, the column map, the parser, row filtering or the split seed) creates a new data version (v1, v2, …) in its own folder, which is never overwritten. Each version has a manifest (file hashes, row and label counts, column map, cleaning rules) and a step-by-step cleaning log listing, for every step, the rows before and after, rows removed or changed (overall and per class), the rule and its reason. Steps that change nothing and decisions not to do something are logged too. Every run records the data version and its hash. After a version change, the current best model is rerun as its own experiment so the effect of the change is measured. [Data versions: see Section 2 and the data changelog.]

## 3.3 Splits and evaluation protocol

For each task and timeout the cleaned data is split once into 70% training, 15% validation and 15% test sets, stratified by class, with random seed 42. The row indices are saved inside the data version, so every run on that version uses exactly the same rows.

- Hyperparameters are chosen by 5-fold stratified cross-validation inside the training set only.
- Candidate models are compared on the validation set.
- The chosen configuration is retrained on training + validation data and scored **once** on the test set, at the final milestone, with 95% confidence intervals from 1,000 bootstrap resamples. A comparison set (the best validation variant of each method type per task, Section 3.8) is fixed before the test set is opened and scored the same way, once; it never changes the chosen configuration. A difference between two models of a task is tested with a paired bootstrap on the same test rows, Holm-corrected for the number of comparisons.
- Scaling, transforms and resampling are steps inside the pipeline, so they are fitted on training folds only and never see validation or test rows.

- Every random step (split, models, tuning sampler, oversampling, bootstrap) uses seed 42. Only the learning-curve and seed-stability experiments (E13, E14) use seeds 42–51, on purpose.

**Paper benchmark.** To compare with Draper-Gil et al. [1], the decision tree and KNN are also run with the paper's own protocol: 10-fold cross-validation on the full file with Weka's default settings (KNN with k = 1; C4.5 with pruning confidence 0.25 and at least 2 samples per leaf). These results are reported in their own table. scikit-learn's decision tree with the entropy criterion is a close substitute for C4.5 [11], not an exact copy, and is called a C4.5 approximation.

**Session-aware check.** Flows from one recording session may fall on both sides of a random split and inflate scores. A group-aware split would keep each session on one side, but the ARFF files hold only the 23 timing features and the label: no capture-file name, address, port or timestamp identifies a session (E00). E11 therefore cannot run, and this is listed as a limitation (Section 7).

## 3.4 Models

| Model | Settings | Experiment |
|---|---|---|
| Majority class | Always predicts the most frequent class (the floor) | E01 |
| Decision tree | Entropy criterion (C4.5 approximation); then tuned | E02, E17 |
| KNN | k = 5, standard scaling; signed log + standard or robust scaling; then tuned | E03, E04, E07 |
| Logistic Regression | Standard scaling; then tuned, signed log optional | E03, E17 |
| SVM, RBF kernel | Standard scaling; then tuned, signed log optional | E03, E17 |
| Random Forest | 300 trees, default settings; then tuned | E05, E07, E17 |
| XGBoost, LightGBM | Default settings; then tuned | E06, E07, E17 |
| Class balancing | Class weights, random oversampling, SMOTE (inside the pipeline) | E08 |
| Two-step model (task B) | VPN / non-VPN model, then one 7-class model per group | E10 |

Every baseline is run on all tasks (A1; A2-VPN and A2-NonVPN as separate models; B) at all four timeouts. Improvement experiments change one thing at a time against the current best model: preprocessing for KNN (E04), tuning with Optuna, 50 trials per model with the same budget for the best baseline (E07), class imbalance handling (E08: none, class weights, random oversampling, SMOTE [10]), feature-group ablation (E09), direct vs two-step classification for task B (E10), and the same 50-trial tuning budget for every remaining method type (E17). The chosen models are then described by a learning curve (E13) and a split-seed stability check (E14), and a published result on a copy of this dataset is reproduced and corrected (E16). Each experiment states its hypothesis and a numeric pass/fail threshold before it is run.

## 3.5 Metrics

The main metric is **macro F1**, which weights every class equally, so small classes count as much as large ones. We also report per-class precision, recall and F1; weighted F1; accuracy; balanced accuracy; confusion matrices and the five most frequent confusion pairs; ROC-AUC (task A1); cross-validation mean and standard deviation; the gap between cross-validation and held-out scores; training time, prediction time per 1,000 flows, model size and peak memory; and, for the final models, Expected Calibration Error with a reliability diagram (E15). Comparisons with the paper use precision against precision and recall against recall.

## 3.6 Tracking and reproducibility

Every run is logged automatically to MLflow with: the experiment ID, the person and machine that ran it (operating system, CPU, memory), the git commit, whether there were uncommitted changes, the hash of the dependency lock file, the configuration file, the data version and hash, all settings, all metrics, timing, model size and memory, plots, and the trained model file (every official run except individual tuning trials). Official runs are made on one machine, so timings are comparable, and only from code that is committed and pushed to the shared repository; the run script refuses otherwise. Each experiment also has a written entry

(hypothesis, threshold, change, result, verdict, decision), and milestones are marked with git tags (v0.1-baseline, v0.2-tuned, v1.0-final).

## 3.7 Preprocessing for distance-based models: the signed log

KNN, logistic regression and the RBF-kernel SVM depend on distances or linear scales, and the raw timing features span six or more orders of magnitude (microseconds up to about 1.2e8), so a few very large values dominate. A log compresses them, but plain log1p is not usable here: the features hold -1 codes meaning "not measurable" (log1p(-1) is minus infinity) and 586 negative minimum inter-arrival times (E00). We therefore apply the **signed log** sign(x)·log1p(|x|) as a step inside the pipeline, before scaling. It is defined for every value, keeps the order of values, keeps -1 apart from 0, and is a fixed formula: nothing is learned from data, so it cannot leak information from validation or test rows. Data v1 is not changed. Alternatives not chosen: shifting each column so its minimum is 0 before the log (a shift learned from the training data, different per column, and -1 merges into small values); logging only the non-negative columns (inconsistent, and large values stay uncompressed); replacing -1 by 0 plus indicator columns (changes the feature set, so E04 would change two things at once). E04 compares the signed log with StandardScaler and with RobustScaler; in E17 logistic regression and the SVM choose it inside their search (both chose it in every task). Tree models are not affected by any order-keeping transform, so they get none.

## 3.8 Phase 3 rules and decisions

These rules were fixed before the Phase 3 runs (28 Oct) unless stated otherwise; each experiment entry repeats the ones it uses before it runs.

- **One change at a time.** Each experiment changes one thing against the **current best** model of each task: the highest 15 s validation macro F1 so far, whatever the margin.
- **Winner rule.** Every experiment runs first at 15 s for all four tasks. A model whose 15 s validation macro F1 beats the task's current best by at least 0.03 is rerun at 30, 60 and 120 s with its 15 s settings (nothing is re-tuned per timeout). Why 0.03: it is above the largest Phase 2 cross-validation standard deviation (0.025), so a smaller gain could be noise from the split; a fixed number set before running keeps the choice honest. Not chosen: rerunning the top two per task, or running everything at every timeout (about four times the runs, and no screen-then-confirm design).
- **Tuning (E07, E17).** Optuna with the TPE sampler (seed 42), 50 trials per search, trials run one after another (parallel trials are not reproducible). Each trial is scored by 5-fold cross-validation macro F1 on the training rows only. E07 tunes, per task, the best two of Random Forest, XGBoost and LightGBM plus KNN with the same budget, so no model wins only because it was tuned more. E17 (added 28 Oct after E10) gives the same budget to every other method type, so the comparison of method types is between tuned models. Search ranges were written into the configurations before running. The best trial is retrained and scored like any other run; trials keep their settings and scores only.
- **Balancing (E08).** Inside the pipeline (imbalanced-learn), so only training folds are resampled. XGBoost has no class-weight option, so its "balanced" step uses per-row sample weights. SMOTE uses 5 neighbours; the smallest training class in any task file has 53 rows (MAIL, A2-NonVPN 30 s), about 42 per cross-validation fold, so it is always possible.
- **Two-step model (E10).** Both steps train on task B's training rows only; the VPN / non-VPN label comes from the B label. The A1 split is never used, because A1 and B hold the same flows split differently, so an A1-trained model would have seen B validation flows.
- **Analysis experiments (E09, E13, E14)** describe the chosen models and change nothing. If a feature set in E09 had scored at least 0.03 above all 23 features, the work would have stopped for a decision, because choosing features on validation scores risks overfitting the choice. E13 and E14 use other seeds on purpose and never touch the seed-42 test rows.
- **Reproducing Isnan et al. (E16)** uses its own data version (kaggle-k1, built by a committed script with its own manifest and hashes); data v1 is untouched, and no E16 result feeds model selection.

- **Cost and resources.** Every model runs on one thread (XGBoost and LightGBM included), with the five cross-validation folds in parallel. This keeps results deterministic and timings comparable with Phase 2. Peak memory is measured twice: with Python's tracemalloc (comparable with Phase 2) and as the peak memory of a fresh process doing the fit, which also sees the native memory of XGBoost and LightGBM. Every model is saved with joblib compression level 3. This is lossless and cuts a 300-tree Random Forest from 171.9 MiB to 22.4 MiB. Both the compressed and the uncompressed size are logged.
- **Final choice (decided 28 Oct after E10, a deviation from the rule set before Phase 3).** The rule fixed before Phase 3 treated any gap below 0.03 as a tie, broken in favour of the simpler model. Applied as written, it would have chosen KNN or the SVM at 0.0146-0.0286 below the best model. The cross-validation standard deviations of the chosen models (0.0028-0.0058) are far below 0.03, and the goal is the best score on unseen traffic. The final configuration per task is therefore the highest validation macro F1 among all model-selection runs (any experiment, any timeout). The only fallback is a saved model file above 50 MB (GitHub's warning size), which none of the chosen models reaches. The old rule's picks are scored on the test set too (E12), so the cost of either rule is visible.
- **Comparison set.** Fixed in `configs/final/compare/` before the test set is opened: per task, the best 15 s validation variant of each of 7 method types (decision tree, KNN, logistic regression, SVM-RBF, Random Forest, XGBoost, LightGBM), 28 models. A type's variants are its default, tuned and balanced runs; KNN includes the E03 and E04 preprocessings; the E10 two-step model counts as XGBoost; the majority baseline is a reference, not a member. Only the four headline models are saved in the repository (`models/final/`); the others stay in the run store.

# 4 Experiments

Each experiment states a hypothesis and a numeric pass/fail threshold before it runs, changes one thing, and ends with a verdict (supported / not supported). Unless stated otherwise, scores are on the **validation** set of the fixed 70/15/15 split (seed 42); models are chosen with 5-fold cross-validation on the training set; the test set is not used until the final experiment (E12).

## 4.0 E00: data findings and data version v1

E00 checked the dataset before any model was trained: the corrected column names, the file-family decision, the odd-value and duplicate rules, the session question and the Kaggle finding. It produced data version v1 (167,909 flows in 16 task files). Section 2 reports it in full.

## 4.1 E01: majority-class baseline

**What.** A model that always predicts the largest training class (`DummyClassifier`), for every task and timeout (16 runs). **Why.** It is the floor every model must beat, and its scores are known in advance, so it also tests the metrics code: accuracy must equal the largest class's share p of the validation set, and macro F1 must equal 2p / ((1 + p) k) for k classes. **Result.** Supported: all 16 runs match both values to 4 decimals (threshold 0.001). The floor in macro F1 is 0.34-0.36 for A1, 0.06-0.10 for A2 and 0.02-0.03 for B. Accuracy alone would look much better (up to 0.56 for A1 and 0.51 for A2-NonVPN 120 s), which is why macro F1 is the main score.

| task | timeout | classes | largest_class | p | val_accuracy | expected_macro_f1 | val_macro_f1 | ok |
|---|---|---|---|---|---|---|---|---|
| A1 | 15 s | 2 | VPN | 0.5185 | 0.5185 | 0.3414 | 0.3414 | yes |
| A1 | 30 s | 2 | VPN | 0.5083 | 0.5083 | 0.337 | 0.337 | yes |
| A1 | 60 s | 2 | Non-VPN | 0.5577 | 0.5577 | 0.358 | 0.358 | yes |
| A1 | 120 s | 2 | Non-VPN | 0.5185 | 0.5185 | 0.3415 | 0.3415 | yes |
| A2-VPN | 15 s | 7 | VPN-BROWSING | 0.2665 | 0.2665 | 0.0601 | 0.0601 | yes |
| A2-VPN | 30 s | 7 | VPN-BROWSING | 0.3599 | 0.3599 | 0.0756 | 0.0756 | yes |
| A2-VPN | 60 s | 7 | VPN-BROWSING | 0.3769 | 0.3769 | 0.0782 | 0.0782 | yes |
| A2-VPN | 120 s | 7 | VPN-BROWSING | 0.4828 | 0.4828 | 0.093 | 0.093 | yes |
| A2-NonVPN | 15 s | 7 | VOIP | 0.3244 | 0.3244 | 0.07 | 0.07 | yes |
| A2-NonVPN | 30 s | 7 | BROWSING | 0.3735 | 0.3735 | 0.0777 | 0.0777 | yes |
| A2-NonVPN | 60 s | 7 | BROWSING | 0.3046 | 0.3046 | 0.0667 | 0.0667 | yes |
| A2-NonVPN | 120 s | 7 | BROWSING | 0.5061 | 0.5061 | 0.096 | 0.096 | yes |
| B | 15 s | 14 | VOIP | 0.156 | 0.156 | 0.0193 | 0.0193 | yes |
| B | 30 s | 14 | BROWSING | 0.1833 | 0.1833 | 0.0221 | 0.0221 | yes |
| B | 60 s | 14 | BROWSING | 0.1697 | 0.1697 | 0.0207 | 0.0207 | yes |
| B | 120 s | 14 | BROWSING | 0.2606 | 0.2606 | 0.0295 | 0.0295 | yes |

## 4.2 E02 and E03: decision tree, KNN, Logistic Regression and SVM-RBF

**What.** Four baselines, each a scikit-learn pipeline, run on all 16 task/timeout files (64 runs): an unpruned entropy decision tree (E02; the closest scikit-learn model to C4.5, so a "C4.5 approximation"), KNN with k = 5 and standard scaling, Logistic Regression and an RBF-kernel SVM, both with standard scaling (E03). Scaling is a pipeline step, fitted on training rows only. **Why.** The tree and KNN are the paper's two algorithms. Logistic Regression and SVM-RBF match the baselines of Isnan et al. [13]. Together they set the bar for the improvements in Phase 3. **Hypotheses and thresholds (written before running).** E02: the tree beats the floor by at least 0.30 in every config and reaches at least 0.80 on A1 at every timeout. E03: all three beat the floor by at least 0.10; KNN is within 0.10 of the tree in at least 12 of 16 configs; Logistic Regression and SVM-

RBF score below the tree in at least 12 of 16 configs, because the features are heavy-tailed (microsecond times up to about 10^8 next to -1 and 0 codes). **Result.** All supported. The tree is the best baseline in all 16 configs (macro F1 0.706-0.890), 0.499-0.765 above the floor. KNN is second everywhere, 0.039-0.101 below the tree (within 0.10 in 15 of 16). Logistic Regression (0.270-0.598) and SVM-RBF (0.274-0.621) are below the tree in all 16.

Validation macro F1 (E01-E03):

| task | timeout | majority | tree | knn | logreg | svm_rbf |
|---|---|---|---|---|---|---|
| A1 | 15 s | 0.341 | 0.890 | 0.813 | 0.585 | 0.602 |
| A1 | 30 s | 0.337 | 0.864 | 0.825 | 0.598 | 0.610 |
| A1 | 60 s | 0.358 | 0.857 | 0.813 | 0.541 | 0.545 |
| A1 | 120 s | 0.341 | 0.857 | 0.807 | 0.576 | 0.621 |
| A2-VPN | 15 s | 0.060 | 0.825 | 0.774 | 0.432 | 0.468 |
| A2-VPN | 30 s | 0.076 | 0.828 | 0.788 | 0.536 | 0.542 |
| A2-VPN | 60 s | 0.078 | 0.775 | 0.730 | 0.339 | 0.330 |
| A2-VPN | 120 s | 0.093 | 0.780 | 0.727 | 0.332 | 0.332 |
| A2-NonVPN | 15 s | 0.070 | 0.829 | 0.773 | 0.449 | 0.487 |
| A2-NonVPN | 30 s | 0.078 | 0.828 | 0.783 | 0.486 | 0.468 |
| A2-NonVPN | 60 s | 0.067 | 0.825 | 0.750 | 0.447 | 0.441 |
| A2-NonVPN | 120 s | 0.096 | 0.839 | 0.769 | 0.482 | 0.471 |
| B | 15 s | 0.019 | 0.759 | 0.660 | 0.307 | 0.296 |
| B | 30 s | 0.022 | 0.748 | 0.647 | 0.320 | 0.311 |
| B | 60 s | 0.021 | 0.706 | 0.617 | 0.271 | 0.278 |
| B | 120 s | 0.030 | 0.716 | 0.637 | 0.270 | 0.274 |

5-fold cross-validation on the training set, mean +- standard deviation:

| task | timeout | majority | tree | knn | logreg | svm_rbf |
|---|---|---|---|---|---|---|
| A1 | 15 s | 0.341 +- 0.000 | 0.888 +- 0.005 | 0.821 +- 0.002 | 0.600 +- 0.011 | 0.608 +- 0.005 |
| A1 | 30 s | 0.337 +- 0.000 | 0.873 +- 0.006 | 0.814 +- 0.010 | 0.600 +- 0.012 | 0.619 +- 0.023 |
| A1 | 60 s | 0.358 +- 0.000 | 0.852 +- 0.010 | 0.808 +- 0.009 | 0.551 +- 0.008 | 0.552 +- 0.006 |
| A1 | 120 s | 0.341 +- 0.000 | 0.857 +- 0.009 | 0.803 +- 0.002 | 0.569 +- 0.008 | 0.608 +- 0.011 |
| A2-VPN | 15 s | 0.060 +- 0.000 | 0.810 +- 0.007 | 0.767 +- 0.008 | 0.410 +- 0.014 | 0.448 +- 0.024 |
| A2-VPN | 30 s | 0.076 +- 0.000 | 0.796 +- 0.009 | 0.769 +- 0.025 | 0.495 +- 0.016 | 0.507 +- 0.021 |
| A2-VPN | 60 s | 0.078 +- 0.000 | 0.760 +- 0.016 | 0.708 +- 0.019 | 0.334 +- 0.022 | 0.331 +- 0.021 |
| A2-VPN | 120 s | 0.093 +- 0.000 | 0.725 +- 0.023 | 0.692 +- 0.016 | 0.343 +- 0.019 | 0.317 +- 0.021 |
| A2-NonVPN | 15 s | 0.070 +- 0.000 | 0.862 +- 0.009 | 0.792 +- 0.010 | 0.459 +- 0.013 | 0.479 +- 0.018 |
| A2-NonVPN | 30 s | 0.078 +- 0.000 | 0.817 +- 0.015 | 0.783 +- 0.017 | 0.459 +- 0.007 | 0.426 +- 0.008 |
| A2-NonVPN | 60 s | 0.067 +- 0.000 | 0.829 +- 0.016 | 0.740 +- 0.013 | 0.454 +- 0.015 | 0.432 +- 0.022 |
| A2-NonVPN | 120 s | 0.096 +- 0.000 | 0.807 +- 0.015 | 0.731 +- 0.013 | 0.472 +- 0.016 | 0.441 +- 0.023 |
| B | 15 s | 0.019 +- 0.000 | 0.757 +- 0.013 | 0.663 +- 0.008 | 0.299 +- 0.010 | 0.278 +- 0.008 |
| B | 30 s | 0.022 +- 0.000 | 0.733 +- 0.008 | 0.654 +- 0.016 | 0.323 +- 0.009 | 0.292 +- 0.008 |
| B | 60 s | 0.021 +- 0.000 | 0.703 +- 0.013 | 0.607 +- 0.010 | 0.275 +- 0.002 | 0.261 +- 0.009 |
| B | 120 s | 0.030 +- 0.000 | 0.684 +- 0.020 | 0.607 +- 0.015 | 0.250 +- 0.005 | 0.251 +- 0.017 |

What the tables show:

- **Cross-validation predicts validation.** For every run, |CV mean - validation| is at most 0.054, and CV standard deviations are at most 0.025. A gap of several points between two models is therefore larger than fold-to-fold noise.
- **Small classes.** Logistic Regression and SVM-RBF score F1 = 0.00 on MAIL / VPN-MAIL in most A2 and B configs. So their macro F1 is far below their weighted F1 (gap up to 0.225); the tree's gap is 0.000-0.069. The SVM keeps 55-88% of the training rows as support vectors, a sign that the RBF kernel separates the standard-scaled features poorly. A log transform or a robust scaler is tested in E04.
- **Easiest and hardest classes.** VoIP is the easiest type (tree F1 0.92-0.99 for VOIP and VPN-VOIP). The weakest are CHAT, VPN-CHAT, STREAMING, MAIL and VPN-MAIL (tree F1 0.42-0.71).
- **Most frequent confusion.** In task B the largest confusion for the tree and KNN, at every timeout, is BROWSING vs VPN-BROWSING in both directions (tree: 29-45 validation flows each way). The model recognises browsing, but not whether it went through the VPN.
- **Timeouts.** B and A2-VPN score lower at 60 s and 120 s. The class mix differs between timeouts (Section 2.8), and the 60 s files hold differently shaped VOIP and FT flows (E00 addendum), so this is not read as an effect of the timeout.
- **Cost.** All baselines train in under 3 s. To predict 1,000 flows, the tree needs 0.30 ms (median), KNN 8.0 ms and the SVM 210 ms. Model files range from 1.4 KB (majority) and 4-5 KB (Logistic Regression) to 4.2 MB (SVM).

Per-class extremes for the tree (E02): macro vs weighted F1, their gap, the generalisation gap, and the best and worst class by F1 and by recall:

| task | timeout | macro F1 | weighted F1 | gap | CV - val | best F1 | worst F1 | best recall | worst recall |
|---|---|---|---|---|---|---|---|---|---|
| A1 | 15 s | 0.89 | 0.89 | 0.0 | -0.002 | VPN 0.89 | Non-VPN 0.89 | Non-VPN 0.89 | VPN 0.89 |
| A1 | 30 s | 0.864 | 0.864 | 0.0 | 0.009 | VPN 0.87 | Non-VPN 0.86 | VPN 0.87 | Non-VPN 0.86 |
| A1 | 60 s | 0.857 | 0.859 | 0.002 | -0.005 | Non-VPN 0.87 | VPN 0.84 | Non-VPN 0.87 | VPN 0.85 |
| A1 | 120 s | 0.857 | 0.857 | 0.0 | 0.0 | Non-VPN 0.86 | VPN 0.85 | Non-VPN 0.86 | VPN 0.85 |
| A2-VPN | 15 s | 0.825 | 0.849 | 0.024 | -0.014 | VPN-VOIP 0.99 | VPN-CHAT 0.65 | VPN-VOIP 0.98 | VPN-CHAT 0.66 |
| A2-VPN | 30 s | 0.828 | 0.847 | 0.02 | -0.031 | VPN-VOIP 0.98 | VPN-CHAT 0.67 | VPN-STREAMING 0.98 | VPN-CHAT 0.70 |
| A2-VPN | 60 s | 0.775 | 0.831 | 0.056 | -0.014 | VPN-VOIP 0.92 | VPN-CHAT 0.63 | VPN-VOIP 0.92 | VPN-FT 0.65 |
| A2-VPN | 120 s | 0.78 | 0.825 | 0.045 | -0.054 | VPN-VOIP 0.98 | VPN-CHAT 0.61 | VPN-VOIP 0.98 | VPN-CHAT 0.57 |
| A2-NonVPN | 15 s | 0.829 | 0.891 | 0.061 | 0.033 | VOIP 0.99 | STREAMING 0.71 | VOIP 0.99 | STREAMING 0.70 |

| task | timeout | macro F1 | weighted F1 | gap | CV - val | best F1 | worst F1 | best recall | worst recall |
|---|---|---|---|---|---|---|---|---|---|
| A2-NonVPN | 30 s | 0.828 | 0.889 | 0.061 | -0.011 | VOIP 0.98 | STREAMING 0.68 | VOIP 0.98 | STREAMING 0.57 |
| A2-NonVPN | 60 s | 0.825 | 0.878 | 0.053 | 0.004 | VOIP 0.97 | STREAMING 0.58 | VOIP 0.97 | STREAMING 0.55 |
| A2-NonVPN | 120 s | 0.839 | 0.893 | 0.054 | -0.033 | VOIP 0.95 | CHAT 0.56 | BROWSING 0.95 | CHAT 0.56 |
| B | 15 s | 0.759 | 0.81 | 0.051 | -0.001 | VOIP 0.98 | VPN-MAIL 0.55 | VOIP 0.99 | VPN-MAIL 0.52 |
| B | 30 s | 0.748 | 0.8 | 0.053 | -0.014 | VPN-VOIP 0.98 | MAIL 0.55 | VPN-VOIP 0.99 | MAIL 0.55 |
| B | 60 s | 0.706 | 0.775 | 0.069 | -0.002 | VOIP 0.92 | VPN-CHAT 0.42 | VOIP 0.92 | VPN-CHAT 0.42 |
| B | 120 s | 0.716 | 0.767 | 0.051 | -0.031 | VOIP 0.97 | STREAMING 0.47 | VOIP 1.00 | CHAT 0.42 |

Top 5 confusion pairs of the tree on the validation set (E02):

| task | timeout | top 5 confusion pairs (true -> predicted: flows) |
|---|---|---|
| A1 | 15 s | VPN -> Non-VPN: 158; Non-VPN -> VPN: 139 |
| A1 | 30 s | Non-VPN -> VPN: 142; VPN -> Non-VPN: 135 |
| A1 | 60 s | Non-VPN -> VPN: 161; VPN -> Non-VPN: 150 |
| A1 | 120 s | Non-VPN -> VPN: 104; VPN -> Non-VPN: 100 |
| A2-VPN | 15 s | VPN-FT -> VPN-CHAT: 28; VPN-CHAT -> VPN-BROWSING: 26; VPN-BROWSING -> VPN-CHAT: 25; VPN-CHAT -> VPN-FT: 20; VPN-FT -> VPN-BROWSING: 19 |
| A2-VPN | 30 s | VPN-BROWSING -> VPN-CHAT: 26; VPN-FT -> VPN-BROWSING: 22; VPN-BROWSING -> VPN-FT: 15; VPN-CHAT -> VPN-BROWSING: 12; VPN-CHAT -> VPN-FT: 12 |
| A2-VPN | 60 s | VPN-FT -> VPN-BROWSING: 17; VPN-BROWSING -> VPN-FT: 16; VPN-FT -> VPN-CHAT: 14; VPN-BROWSING -> VPN-VOIP: 13; VPN-CHAT -> VPN-FT: 12 |
| A2-VPN | 120 s | VPN-FT -> VPN-BROWSING: 18; VPN-BROWSING -> VPN-FT: 16; VPN-CHAT -> VPN-BROWSING: 11; VPN-FT -> VPN-P2P: 9; VPN-P2P -> VPN-BROWSING: 9 |
| A2-NonVPN | 15 s | BROWSING -> CHAT: 18; CHAT -> BROWSING: 15; BROWSING -> STREAMING: 12; FT -> BROWSING: 12; FT -> CHAT: 11 |
| A2-NonVPN | 30 s | CHAT -> BROWSING: 12; STREAMING -> BROWSING: 11; BROWSING -> CHAT: 10; BROWSING -> FT: 10; FT -> CHAT: 9 |
| A2-NonVPN | 60 s | CHAT -> BROWSING: 20; BROWSING -> CHAT: 19; MAIL -> FT: 8; FT -> BROWSING: 7; BROWSING -> MAIL: 6 |
| A2-NonVPN | 120 s | CHAT -> BROWSING: 12; P2P -> BROWSING: 8; FT -> BROWSING: 7; BROWSING -> P2P: 6; BROWSING -> CHAT: 5 |
| B | 15 s | VPN-BROWSING -> BROWSING: 41; BROWSING -> VPN-BROWSING: 39; VPN-CHAT -> VPN-FT: 25; VPN-FT -> VPN-CHAT: 24; VPN-BROWSING -> VPN-CHAT: 18 |
| B | 30 s | VPN-BROWSING -> BROWSING: 44; BROWSING -> VPN-BROWSING: 36; VPN-BROWSING -> VPN-CHAT: 13; VPN-FT -> VPN-CHAT: 12; VPN-CHAT -> VPN-BROWSING: 11 |
| B | 60 s | BROWSING -> VPN-BROWSING: 37; VPN-BROWSING -> BROWSING: 29; VPN-MAIL -> MAIL: 15; VPN-P2P -> P2P: 14; VPN-VOIP -> VOIP: 14 |
| B | 120 s | BROWSING -> VPN-BROWSING: 45; VPN-BROWSING -> BROWSING: 32; VPN-BROWSING -> VPN-FT: 17; VPN-FT -> VPN-BROWSING: 12; VPN-BROWSING -> VPN-P2P: 10 |

Model complexity (E02 tree, E03 SVM):

| task | timeout | n_train | tree depth | tree nodes | tree leaves | SVM support vectors |
|---|---|---|---|---|---|---|
| A1 | 15 s | 12,626 | 27 | 1,789 | 895 | 10,292 |
| A1 | 30 s | 9,526 | 35 | 1,601 | 801 | 7,976 |
| A1 | 60 s | 10,273 | 27 | 1,917 | 959 | 7,882 |

| task | timeout | n_train | tree depth | tree nodes | tree leaves | SVM support vectors |
|---|---|---|---|---|---|---|
| A1 | 120 s | 6,666 | 25 | 1,243 | 622 | 5,501 |
| A2-VPN | 15 s | 6,566 | 23 | 1,407 | 704 | 4,497 |
| A2-VPN | 30 s | 4,860 | 20 | 1,067 | 534 | 3,108 |
| A2-VPN | 60 s | 4,565 | 24 | 1,249 | 625 | 3,265 |
| A2-VPN | 120 s | 3,252 | 21 | 985 | 493 | 2,368 |
| A2-NonVPN | 15 s | 6,085 | 16 | 743 | 372 | 3,329 |
| A2-NonVPN | 30 s | 4,685 | 19 | 751 | 376 | 2,751 |
| A2-NonVPN | 60 s | 5,744 | 24 | 1,001 | 501 | 3,803 |
| A2-NonVPN | 120 s | 3,458 | 18 | 591 | 296 | 1,907 |
| B | 15 s | 12,651 | 30 | 3,409 | 1,705 | 11,076 |
| B | 30 s | 9,546 | 23 | 2,901 | 1,451 | 8,368 |
| B | 60 s | 10,309 | 26 | 3,437 | 1,719 | 8,796 |
| B | 120 s | 6,711 | 22 | 2,481 | 1,241 | 5,813 |

Cost per model over the 16 configs:

| model | train time (s) median | train time (s) max | 5-fold CV time (s) median | 5-fold CV time (s) max | predict ms / 1,000 flows median | predict ms / 1,000 flows max | model file (KB) median | model file (KB) max | peak fit memory (MB) median | peak fit memory (MB) max |
|---|---|---|---|---|---|---|---|---|---|---|
| majority | 0.004 | 0.008 | 0.157 | 2.854 | 0.091 | 0.18 | 1.414 | 1.523 | 0.623 | 1.182 |
| tree | 0.112 | 0.304 | 0.293 | 0.652 | 0.298 | 0.531 | 134.49 | 592.916 | 1.257 | 2.392 |
| knn | 0.01 | 0.021 | 0.182 | 0.281 | 8.031 | 13.338 | 1242.967 | 2374.631 | 2.999 | 5.722 |
| logreg | 0.084 | 0.428 | 0.212 | 0.603 | 0.378 | 0.69 | 3.829 | 5.196 | 3.433 | 8.144 |
| svm_rbf | 0.626 | 2.913 | 1.61 | 3.784 | 209.808 | 596.678 | 1175.091 | 4288.896 | 3.552 | 7.781 |

Peak fit memory is measured with Python's tracemalloc, which counts NumPy arrays but not memory allocated inside compiled libraries (for example libsvm), so SVM memory is a lower bound. Training time comes from a fit without tracemalloc and peak memory from a separate fit of a clone. The first Phase 2 runs (commits 1817c5a and 36c56a1) timed the fit with tracemalloc on, which slowed it (x1.05 tree, x1.00 SVM, x1.19 Logistic Regression, x2.3 KNN in a recheck); all 144 configs were rerun from the fixed code (commit c450c3e) with identical scores, and these tables use the rerun. The first runs also logged process_max_rss_mb, a high-water mark of the whole batch process rather than a per-run figure; it is not used.

The same tables for KNN, Logistic Regression and SVM-RBF, plus accuracy, balanced accuracy, weighted F1 and A1 ROC-AUC for every run, are in `reports/baseline_tables.md` (Appendix).

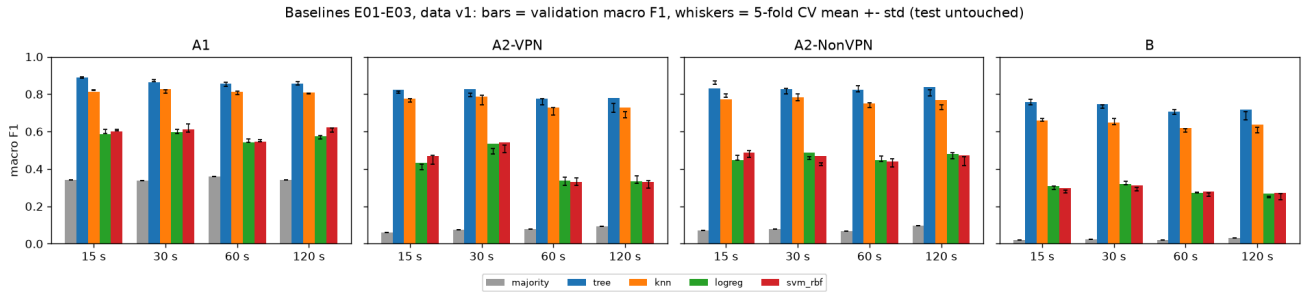



*Figure 4. Baselines E01-E03*

## 4.3 Paper benchmark (part of E02 and E03)

**What.** The paper's own protocol [1]: 10-fold stratified cross-validation on the whole file, with predictions pooled over the folds as Weka reports them, for its two algorithms with Weka's default settings. J48 (C4.5)

defaults are a pruning confidence of 0.25 and at least 2 instances per leaf; scikit-learn has no C4.5 pruning, so our stand-in is an unpruned entropy tree with at least 2 rows per leaf. IBk (KNN) defaults are k = 1 with a Euclidean distance that normalises each attribute to its range; our stand-in is min-max scaling followed by 1-NN. The defaults were checked in the Weka documentation [16] (also cited in `notebooks/02_baselines.ipynb`). Each algorithm ran on two row sources per file (64 runs): every row as published (duplicates kept, values untouched), which is closest to the paper's input, and data v1 (deduplicated). **Why.** It tests whether we reproduce the paper's results before we try to improve on them. We compare precision with precision and recall with recall, never the paper's precision with our F1. These scores are never used to choose a model. **Hypotheses and thresholds.** On the published rows, at least 12 of the 16 A1 values of the paper's Figure 2 are matched within +-0.05, for each algorithm. Because duplicates can put a copy of a flow in the training folds, removing them lowers 10-fold macro F1 in at least 12 of 16 files. **Result.** Reproduction supported: our C4.5 approximation matches all 16 A1 values (largest difference 0.021, close to the +-0.01 reading error of the chart), and 1-NN matches 15 of 16 (the miss is VPN recall at 120 s: 0.803 vs 0.860). The duplicate hypothesis is not supported (6 of 16 files lower for the tree, 2 of 16 for 1-NN).

A1, paper (Figure 2, read to +-0.01) vs ours (`diff` = ours - paper):

| algorithm | class | timeout | metric | paper | paper_source | ours_published | ours_v1 | diff_published | diff_v1 |
|---|---|---|---|---|---|---|---|---|---|
| C4.5 | Non-VPN | 15 | precision | 0.906 | chart +-0.01 and text | 0.886 | 0.882 | -0.02 | -0.024 |
| C4.5 | Non-VPN | 30 | precision | 0.885 | chart +-0.01 | 0.866 | 0.866 | -0.019 | -0.019 |
| C4.5 | Non-VPN | 60 | precision | 0.864 | chart +-0.01 | 0.86 | 0.876 | -0.004 | 0.012 |
| C4.5 | Non-VPN | 120 | precision | 0.887 | chart +-0.01 and text | 0.87 | 0.863 | -0.017 | -0.024 |
| C4.5 | Non-VPN | 15 | recall | 0.877 | chart +-0.01 | 0.882 | 0.903 | 0.005 | 0.026 |
| C4.5 | Non-VPN | 30 | recall | 0.846 | chart +-0.01 | 0.857 | 0.886 | 0.011 | 0.04 |
| C4.5 | Non-VPN | 60 | recall | 0.888 | chart +-0.01 | 0.889 | 0.898 | 0.001 | 0.01 |
| C4.5 | Non-VPN | 120 | recall | 0.843 | chart +-0.01 | 0.853 | 0.89 | 0.01 | 0.047 |
| C4.5 | VPN | 15 | precision | 0.89 | chart +-0.01 and text | 0.892 | 0.908 | 0.002 | 0.018 |
| C4.5 | VPN | 30 | precision | 0.868 | chart +-0.01 | 0.874 | 0.888 | 0.006 | 0.02 |
| C4.5 | VPN | 60 | precision | 0.856 | chart +-0.01 | 0.857 | 0.867 | 0.001 | 0.011 |
| C4.5 | VPN | 120 | precision | 0.863 | chart +-0.01 and text | 0.868 | 0.877 | 0.005 | 0.014 |
| C4.5 | VPN | 15 | recall | 0.917 | chart +-0.01 | 0.896 | 0.888 | -0.021 | -0.029 |
| C4.5 | VPN | 30 | recall | 0.901 | chart +-0.01 | 0.881 | 0.867 | -0.02 | -0.034 |
| C4.5 | VPN | 60 | recall | 0.827 | chart +-0.01 | 0.821 | 0.84 | -0.006 | 0.013 |
| C4.5 | VPN | 120 | recall | 0.902 | chart +-0.01 | 0.884 | 0.849 | -0.018 | -0.053 |
| KNN | Non-VPN | 15 | precision | 0.846 | chart +-0.01 and text | 0.821 | 0.846 | -0.025 | 0.0 |
| KNN | Non-VPN | 30 | precision | 0.84 | chart +-0.01 | 0.805 | 0.83 | -0.035 | -0.01 |
| KNN | Non-VPN | 60 | precision | 0.838 | chart +-0.01 | 0.831 | 0.854 | -0.007 | 0.016 |

| algorithm | class | timeout | metric | paper | paper_source | ours_published | ours_v1 | diff_published | diff_v1 |
|---|---|---|---|---|---|---|---|---|---|
| KNN | Non-VPN | 120 | precision | 0.837 | chart +-0.01 and text | 0.794 | 0.83 | -0.043 | -0.007 |
| KNN | Non-VPN | 15 | recall | 0.832 | chart +-0.01 | 0.857 | 0.848 | 0.025 | 0.016 |
| KNN | Non-VPN | 30 | recall | 0.808 | chart +-0.01 | 0.839 | 0.831 | 0.031 | 0.023 |
| KNN | Non-VPN | 60 | recall | 0.851 | chart +-0.01 | 0.857 | 0.849 | 0.006 | -0.002 |
| KNN | Non-VPN | 120 | recall | 0.787 | chart +-0.01 | 0.827 | 0.825 | 0.04 | 0.038 |
| KNN | VPN | 15 | precision | 0.848 | chart +-0.01 and text | 0.863 | 0.859 | 0.015 | 0.011 |
| KNN | VPN | 30 | precision | 0.835 | chart +-0.01 | 0.85 | 0.836 | 0.015 | 0.001 |
| KNN | VPN | 60 | precision | 0.812 | chart +-0.01 | 0.816 | 0.811 | 0.004 | -0.001 |
| KNN | VPN | 120 | precision | 0.815 | chart +-0.01 and text | 0.836 | 0.813 | 0.021 | -0.002 |
| KNN | VPN | 15 | recall | 0.861 | chart +-0.01 | 0.83 | 0.857 | -0.031 | -0.004 |
| KNN | VPN | 30 | recall | 0.862 | chart +-0.01 | 0.818 | 0.835 | -0.044 | -0.027 |
| KNN | VPN | 60 | recall | 0.795 | chart +-0.01 | 0.784 | 0.817 | -0.011 | 0.022 |
| KNN | VPN | 120 | recall | 0.86 | chart +-0.01 | 0.803 | 0.818 | -0.057 | -0.042 |

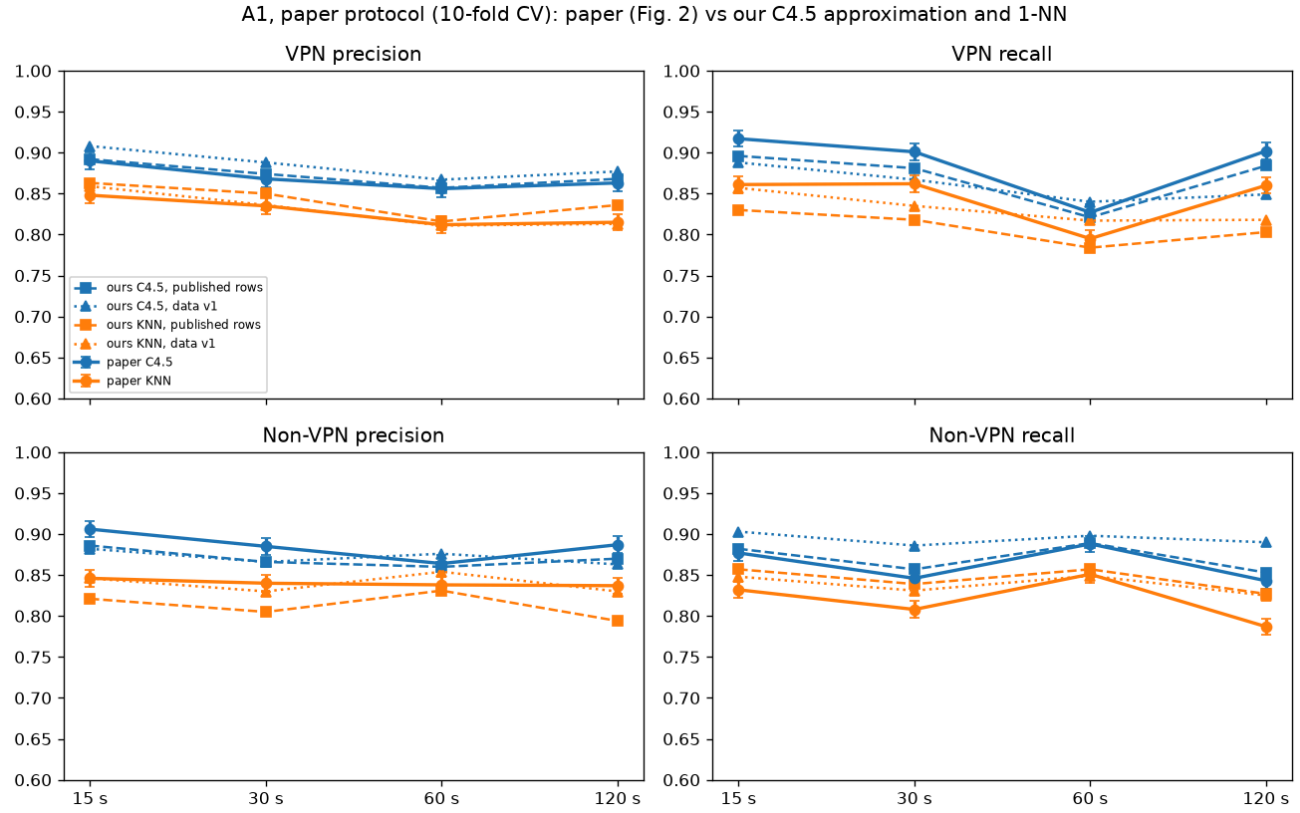



*Figure 5. Paper vs ours, A1*

The plan asks for the comparison as one table per task. The tables below give our 10-fold precision and recall for every class, both algorithms and all timeouts, on the published rows (closest to the paper's input), with the paper's value in brackets wherever the paper gives one (C = read from its chart, +-0.01; T = stated in its text). The same tables on data v1 are in `reports/baseline_tables.md`.

A2-VPN:

| class | metric | C4.5 15 s | C4.5 30 s | C4.5 60 s | C4.5 120 s | KNN 15 s | KNN 30 s | KNN 60 s | KNN 120 s |
|---|---|---|---|---|---|---|---|---|---|
| VPN-BROWSING | precision | 0.872 | 0.878 | 0.873 | 0.878 | 0.830 | 0.874 | 0.881 | 0.795 |
| VPN-BROWSING | recall | 0.884 | 0.916 | 0.918 | 0.906 | 0.846 | 0.893 | 0.902 | 0.888 |
| VPN-CHAT | precision | 0.661 | 0.647 | 0.568 | 0.532 | 0.511 | 0.490 | 0.427 | 0.576 |
| VPN-CHAT | recall | 0.687 | 0.628 | 0.562 | 0.473 | 0.651 | 0.667 | 0.658 | 0.562 |
| VPN-FT | precision | 0.792 | 0.744 | 0.676 | 0.649 | 0.773 | 0.764 | 0.714 | 0.674 |
| VPN-FT | recall | 0.770 | 0.690 | 0.610 | 0.605 | 0.735 | 0.689 | 0.648 | 0.595 |
| VPN-MAIL | precision | 0.808 | 0.891 | 0.778 | 0.864 | 0.766 | 0.912 | 0.753 | 0.859 |
| VPN-MAIL | recall | 0.762 | 0.887 | 0.715 | 0.852 | 0.766 | 0.947 | 0.734 | 0.904 |
| VPN-P2P | precision | 0.827 | 0.831 | 0.793 | 0.810 | 0.736 | 0.806 | 0.727 | 0.771 |
| VPN-P2P | recall | 0.863 | 0.870 | 0.848 | 0.862 | 0.517 | 0.585 | 0.527 | 0.567 |
| VPN-STREAMING | precision | 0.922 | 0.926 | 0.875 | 0.838 | 0.906 | 0.876 | 0.838 | 0.815 |
| VPN-STREAMING | recall | 0.891 | 0.876 | 0.782 | 0.752 | 0.914 | 0.856 | 0.787 | 0.759 |
| VPN-VOIP | precision | 0.990 | 0.992 | 0.939 | 0.973 | 0.990 | 0.993 | 0.911 | 0.984 |
| VPN-VOIP | recall | 0.979 | 0.975 | 0.914 | 0.925 | 0.987 | 0.981 | 0.907 | 0.951 |
| AVERAGE (macro) | precision | 0.839 (paper 0.840 T) | 0.844 | 0.786 | 0.792 | 0.788 | 0.816 | 0.750 | 0.782 |
| AVERAGE (macro) | recall | 0.834 | 0.835 | 0.764 | 0.768 | 0.774 | 0.803 | 0.738 | 0.747 |

A2-NonVPN:

| class | metric | C4.5 15 s | C4.5 30 s | C4.5 60 s | C4.5 120 s | KNN 15 s | KNN 30 s | KNN 60 s | KNN 120 s |
|---|---|---|---|---|---|---|---|---|---|
| BROWSING | precision | 0.904 | 0.915 | 0.897 | 0.928 | 0.866 | 0.883 | 0.865 | 0.922 |
| BROWSING | recall | 0.930 | 0.934 | 0.922 | 0.948 | 0.880 | 0.912 | 0.872 | 0.932 |
| CHAT | precision | 0.813 | 0.769 | 0.738 | 0.656 | 0.558 | 0.508 | 0.483 | 0.331 |
| CHAT | recall | 0.798 | 0.793 | 0.753 | 0.694 | 0.692 | 0.662 | 0.661 | 0.599 |
| FT | precision | 0.810 | 0.794 | 0.793 | 0.796 | 0.812 | 0.769 | 0.847 | 0.791 |
| FT | recall | 0.853 | 0.824 | 0.873 | 0.851 | 0.670 | 0.629 | 0.734 | 0.571 |
| MAIL | precision | 0.849 | 0.754 | 0.866 | 0.859 | 0.709 | 0.734 | 0.762 | 0.709 |
| MAIL | recall | 0.835 | 0.605 | 0.786 | 0.789 | 0.735 | 0.716 | 0.723 | 0.697 |
| P2P | precision | 0.971 | 0.943 | 0.948 | 0.946 | 0.907 | 0.897 | 0.893 | 0.914 |
| P2P | recall | 0.955 | 0.916 | 0.919 | 0.929 | 0.890 | 0.883 | 0.897 | 0.911 |
| STREAMING | precision | 0.837 | 0.769 | 0.757 | 0.768 | 0.782 | 0.759 | 0.655 | 0.657 |
| STREAMING | recall | 0.701 | 0.664 | 0.567 | 0.558 | 0.685 | 0.617 | 0.504 | 0.524 |
| VOIP | precision | 0.992 | 0.990 | 0.973 | 0.989 | 0.990 | 0.980 | 0.968 | 0.931 |
| VOIP | recall | 0.988 | 0.979 | 0.946 | 0.941 | 0.987 | 0.977 | 0.954 | 0.929 |
| AVERAGE (macro) | precision | 0.882 (paper 0.890 T) | 0.848 | 0.853 | 0.849 | 0.803 | 0.790 | 0.782 | 0.751 |

| class | metric | C4.5 15 s | C4.5 30 s | C4.5 60 s | C4.5 120 s | KNN 15 s | KNN 30 s | KNN 60 s | KNN 120 s |
|---|---|---|---|---|---|---|---|---|---|
| AVERAGE (macro) | recall | 0.866 | 0.817 | 0.824 | 0.816 | 0.791 | 0.771 | 0.764 | 0.738 |

B:

| class | metric | C4.5 15 s | C4.5 30 s | C4.5 60 s | C4.5 120 s | KNN 15 s | KNN 30 s | KNN 60 s | KNN 120 s |
|---|---|---|---|---|---|---|---|---|---|
| BROWSING | precision | 0.771 | 0.790 | 0.779 | 0.800 | 0.690 | 0.748 | 0.746 | 0.807 |
| BROWSING | recall | 0.820 | 0.831 | 0.827 | 0.847 | 0.700 | 0.758 | 0.741 | 0.794 |
| CHAT | precision | 0.677 | 0.620 | 0.631 | 0.510 | 0.349 | 0.292 | 0.336 | 0.164 |
| CHAT | recall | 0.727 | 0.667 | 0.667 | 0.554 | 0.567 | 0.536 | 0.586 | 0.488 |
| FT | precision | 0.717 | 0.666 | 0.807 | 0.687 | 0.744 | 0.685 | 0.809 | 0.683 |
| FT | recall | 0.653 | 0.614 | 0.743 | 0.535 | 0.622 | 0.552 | 0.717 | 0.487 |
| MAIL | precision | 0.633 (paper 0.671 T) | 0.617 | 0.741 | 0.680 (paper 0.790 T) | 0.618 | 0.712 | 0.674 | 0.548 |
| MAIL | recall | 0.651 | 0.617 | 0.727 | 0.665 | 0.643 | 0.642 | 0.645 | 0.616 |
| P2P | precision | 0.880 | 0.851 | 0.842 | 0.874 | 0.749 | 0.743 | 0.760 | 0.818 |
| P2P | recall | 0.893 | 0.845 | 0.863 | 0.891 | 0.732 | 0.724 | 0.752 | 0.804 |
| STREAMING | precision | 0.713 | 0.657 | 0.496 | 0.510 | 0.679 | 0.630 | 0.536 | 0.477 |
| STREAMING | recall | 0.670 | 0.623 | 0.472 | 0.495 | 0.629 | 0.532 | 0.440 | 0.394 |
| VOIP | precision | 0.983 | 0.976 | 0.912 | 0.934 | 0.980 | 0.974 | 0.923 | 0.895 |
| VOIP | recall | 0.981 | 0.971 | 0.905 | 0.903 | 0.983 | 0.972 | 0.903 | 0.895 |
| VPN-BROWSING | precision | 0.773 (paper 0.771 T) | 0.800 | 0.797 | 0.807 (paper 0.809 T) | 0.689 (paper 0.691 T) | 0.738 | 0.760 | 0.742 (paper 0.743 T) |
| VPN-BROWSING | recall | 0.768 | 0.778 | 0.802 | 0.799 | 0.711 | 0.759 | 0.780 | 0.703 |
| VPN-CHAT | precision | 0.616 | 0.607 | 0.557 | 0.534 | 0.510 (paper 0.501 T) | 0.503 | 0.460 | 0.466 (paper 0.501 T) |
| VPN-CHAT | recall | 0.571 | 0.560 | 0.477 | 0.444 | 0.507 | 0.471 | 0.442 | 0.476 |
| VPN-FT | precision | 0.759 | 0.717 | 0.639 | 0.661 | 0.693 | 0.664 | 0.584 | 0.549 |
| VPN-FT | recall | 0.712 | 0.661 | 0.532 | 0.539 | 0.637 | 0.593 | 0.528 | 0.580 |
| VPN-MAIL | precision | 0.770 (paper 0.739 T) | 0.828 | 0.702 | 0.804 (paper 0.786 T) | 0.687 (paper 0.688 T) | 0.843 | 0.599 | 0.820 (paper 0.688 T) |
| VPN-MAIL | recall | 0.670 | 0.808 | 0.608 | 0.802 | 0.644 | 0.902 | 0.554 | 0.878 |
| VPN-P2P | precision | 0.615 | 0.620 | 0.547 | 0.629 | 0.591 | 0.666 | 0.593 | 0.669 |
| VPN-P2P | recall | 0.767 | 0.771 | 0.745 | 0.806 | 0.428 | 0.498 | 0.441 | 0.514 |
| VPN-STREAMING | precision | 0.885 | 0.871 | 0.863 | 0.875 | 0.836 | 0.774 | 0.774 | 0.713 |
| VPN-STREAMING | recall | 0.823 | 0.812 | 0.766 | 0.772 | 0.848 | 0.772 | 0.782 | 0.703 |
| VPN-VOIP | precision | 0.986 | 0.980 | 0.893 | 0.942 | 0.978 | 0.963 | 0.807 | 0.921 |
| VPN-VOIP | recall | 0.973 | 0.968 | 0.845 | 0.889 | 0.972 | 0.965 | 0.809 | 0.912 |
| AVERAGE (macro) | precision | 0.770 | 0.757 | 0.729 | 0.732 | 0.700 | 0.710 | 0.669 | 0.662 |
| AVERAGE (macro) | recall | 0.763 | 0.752 | 0.713 | 0.710 | 0.687 | 0.691 | 0.651 | 0.660 |

For A2 and B the paper's Figure 3 draws two unlabelled lines per timeout, which can't be assigned to C4.5 or KNN. So only the values the paper states in its text are compared. The paper doesn't say whether its

"average" is a plain or a class-weighted mean, so both are shown. On the published rows the plain (macro) mean matches it within 0.013 in every case, which suggests the paper used the plain mean.

For averages the paper does not say whether it is a plain or a class-weighted mean (Weka prints the weighted one), so both are shown.

| task | algorithm | class | timeout | metric | paper | rows | ours | ours_kind | diff |
|---|---|---|---|---|---|---|---|---|---|
| A2-VPN | C4.5 | AVERAGE | 15 | precision | 0.84 | published | 0.839 | macro | -0.001 |
| A2-VPN | C4.5 | AVERAGE | 15 | precision | 0.84 | published | 0.853 | weighted | 0.013 |
| A2-VPN | C4.5 | AVERAGE | 15 | precision | 0.84 | v1 | 0.827 | macro | -0.013 |
| A2-VPN | C4.5 | AVERAGE | 15 | precision | 0.84 | v1 | 0.846 | weighted | 0.006 |
| A2-NonVPN | C4.5 | AVERAGE | 15 | precision | 0.89 | published | 0.882 | macro | -0.008 |
| A2-NonVPN | C4.5 | AVERAGE | 15 | precision | 0.89 | published | 0.914 | weighted | 0.024 |
| A2-NonVPN | C4.5 | AVERAGE | 15 | precision | 0.89 | v1 | 0.886 | macro | -0.004 |
| A2-NonVPN | C4.5 | AVERAGE | 15 | precision | 0.89 | v1 | 0.92 | weighted | 0.03 |
| B | C4.5 | VPN-BROWSING | 15 | precision | 0.771 | published | 0.773 | class | 0.002 |
| B | C4.5 | VPN-BROWSING | 15 | precision | 0.771 | v1 | 0.76 | class | -0.011 |
| B | C4.5 | VPN-BROWSING | 120 | precision | 0.809 | published | 0.807 | class | -0.002 |
| B | C4.5 | VPN-BROWSING | 120 | precision | 0.809 | v1 | 0.76 | class | -0.049 |
| B | C4.5 | VPN-MAIL | 15 | precision | 0.739 | published | 0.77 | class | 0.031 |
| B | C4.5 | VPN-MAIL | 15 | precision | 0.739 | v1 | 0.718 | class | -0.021 |
| B | C4.5 | VPN-MAIL | 120 | precision | 0.786 | published | 0.804 | class | 0.018 |
| B | C4.5 | VPN-MAIL | 120 | precision | 0.786 | v1 | 0.682 | class | -0.104 |
| B | C4.5 | MAIL | 15 | precision | 0.671 | published | 0.633 | class | -0.038 |
| B | C4.5 | MAIL | 15 | precision | 0.671 | v1 | 0.625 | class | -0.046 |
| B | C4.5 | MAIL | 120 | precision | 0.79 | published | 0.68 | class | -0.11 |
| B | C4.5 | MAIL | 120 | precision | 0.79 | v1 | 0.613 | class | -0.177 |
| B | KNN | VPN-BROWSING | 15 | precision | 0.691 | published | 0.689 | class | -0.002 |
| B | KNN | VPN-BROWSING | 15 | precision | 0.691 | v1 | 0.695 | class | 0.004 |
| B | KNN | VPN-BROWSING | 120 | precision | 0.743 | published | 0.742 | class | -0.001 |
| B | KNN | VPN-BROWSING | 120 | precision | 0.743 | v1 | 0.7 | class | -0.043 |
| B | KNN | VPN-CHAT | 15 | precision | 0.501 | published | 0.51 | class | 0.009 |
| B | KNN | VPN-CHAT | 15 | precision | 0.501 | v1 | 0.509 | class | 0.008 |
| B | KNN | VPN-CHAT | 120 | precision | 0.501 | published | 0.466 | class | -0.035 |
| B | KNN | VPN-CHAT | 120 | precision | 0.501 | v1 | 0.494 | class | -0.007 |
| B | KNN | VPN-MAIL | 15 | precision | 0.688 | published | 0.687 | class | -0.001 |
| B | KNN | VPN-MAIL | 15 | precision | 0.688 | v1 | 0.633 | class | -0.055 |
| B | KNN | VPN-MAIL | 120 | precision | 0.688 | published | 0.82 | class | 0.132 |
| B | KNN | VPN-MAIL | 120 | precision | 0.688 | v1 | 0.521 | class | -0.167 |
| B | C4.5 | AVERAGE_BEST_TIMEOUT | best | precision | 0.783 | published | 0.77 | macro | -0.013 |
| B | C4.5 | AVERAGE_BEST_TIMEOUT | best | precision | 0.783 | published | 0.808 | weighted | 0.025 |

| task | algorithm | class | timeout | metric | paper | rows | ours | ours_kind | diff |
|---|---|---|---|---|---|---|---|---|---|
| B | C4.5 | AVERAGE_BEST_TIMEOUT | best | precision | 0.783 | v1 | 0.774 | macro | -0.009 |
| B | C4.5 | AVERAGE_BEST_TIMEOUT | best | precision | 0.783 | v1 | 0.815 | weighted | 0.032 |
| B | KNN | AVERAGE_BEST_TIMEOUT | best | precision | 0.711 | published | 0.71 | macro | -0.001 |
| B | KNN | AVERAGE_BEST_TIMEOUT | best | precision | 0.711 | published | 0.745 | weighted | 0.034 |
| B | KNN | AVERAGE_BEST_TIMEOUT | best | precision | 0.711 | v1 | 0.707 | macro | -0.004 |
| B | KNN | AVERAGE_BEST_TIMEOUT | best | precision | 0.711 | v1 | 0.755 | weighted | 0.044 |

**Duplicates.** Removing the exact duplicates changed the 10-fold macro F1 by -0.018 to +0.014 for the tree (mean +0.001) and by -0.017 to +0.040 for 1-NN (mean +0.012):

| algo | task | timeout | rows_published | rows_v1 | f1_published | f1_v1 | v1_minus_published |
|---|---|---|---|---|---|---|---|
| j48 | A1 | 15 | 18,758 | 18,038 | 0.8888 | 0.8954 | 0.0067 |
| j48 | A1 | 30 | 14,651 | 13,610 | 0.8696 | 0.8766 | 0.0069 |
| j48 | A1 | 60 | 15,515 | 14,677 | 0.8562 | 0.8702 | 0.014 |
| j48 | A1 | 120 | 10,782 | 9,524 | 0.8688 | 0.8694 | 0.0007 |
| j48 | A2-VPN | 15 | 9,793 | 9,380 | 0.836 | 0.8197 | -0.0162 |
| j48 | A2-VPN | 30 | 7,734 | 6,944 | 0.8389 | 0.821 | -0.0179 |
| j48 | A2-VPN | 60 | 6,935 | 6,523 | 0.7741 | 0.7702 | -0.0039 |
| j48 | A2-VPN | 120 | 5,631 | 4,648 | 0.779 | 0.7719 | -0.0071 |
| j48 | A2-NonVPN | 15 | 8,965 | 8,694 | 0.8731 | 0.8768 | 0.0037 |
| j48 | A2-NonVPN | 30 | 6,917 | 6,694 | 0.8303 | 0.8359 | 0.0055 |
| j48 | A2-NonVPN | 60 | 8,580 | 8,206 | 0.8359 | 0.8483 | 0.0123 |
| j48 | A2-NonVPN | 120 | 5,151 | 4,941 | 0.8294 | 0.8346 | 0.0052 |
| j48 | B | 15 | 18,758 | 18,074 | 0.765 | 0.7707 | 0.0057 |
| j48 | B | 30 | 14,651 | 13,638 | 0.7535 | 0.7488 | -0.0047 |
| j48 | B | 60 | 15,515 | 14,729 | 0.7185 | 0.7293 | 0.0107 |
| j48 | B | 120 | 10,782 | 9,589 | 0.7183 | 0.718 | -0.0003 |
| ibk | A1 | 15 | 18,758 | 18,038 | 0.8424 | 0.8524 | 0.01 |
| ibk | A1 | 30 | 14,651 | 13,610 | 0.8276 | 0.8331 | 0.0055 |
| ibk | A1 | 60 | 15,515 | 14,677 | 0.8217 | 0.8326 | 0.0109 |
| ibk | A1 | 120 | 10,782 | 9,524 | 0.8146 | 0.8218 | 0.0072 |
| ibk | A2-VPN | 15 | 9,793 | 9,380 | 0.7766 | 0.793 | 0.0164 |
| ibk | A2-VPN | 30 | 7,734 | 6,944 | 0.8047 | 0.81 | 0.0054 |
| ibk | A2-VPN | 60 | 6,935 | 6,523 | 0.7377 | 0.7552 | 0.0175 |
| ibk | A2-VPN | 120 | 5,631 | 4,648 | 0.7611 | 0.7441 | -0.017 |
| ibk | A2-NonVPN | 15 | 8,965 | 8,694 | 0.7949 | 0.8118 | 0.0169 |
| ibk | A2-NonVPN | 30 | 6,917 | 6,694 | 0.7769 | 0.7989 | 0.022 |
| ibk | A2-NonVPN | 60 | 8,580 | 8,206 | 0.7687 | 0.7899 | 0.0213 |
| ibk | A2-NonVPN | 120 | 5,151 | 4,941 | 0.7349 | 0.7751 | 0.0402 |
| ibk | B | 15 | 18,758 | 18,074 | 0.69 | 0.7071 | 0.0171 |
| ibk | B | 30 | 14,651 | 13,638 | 0.6958 | 0.6931 | -0.0027 |
| ibk | B | 60 | 15,515 | 14,729 | 0.6561 | 0.6697 | 0.0137 |
| ibk | B | 120 | 10,782 | 9,589 | 0.6531 | 0.661 | 0.0079 |

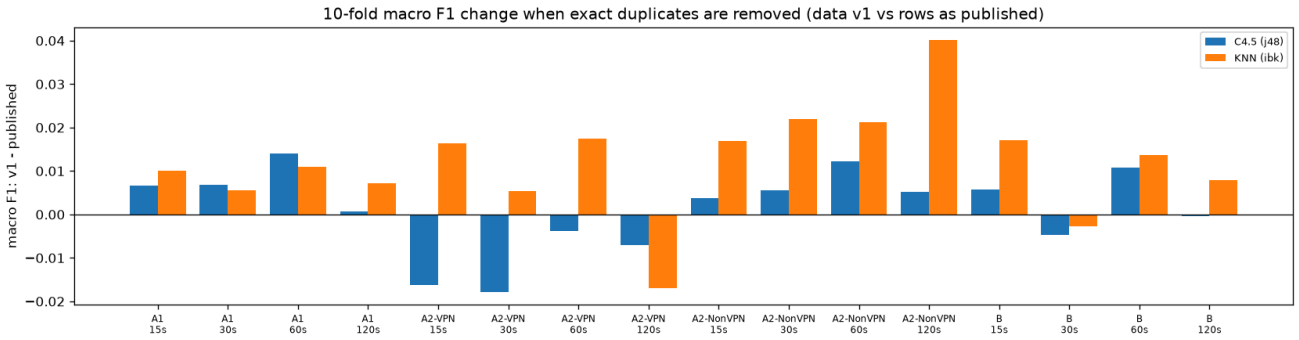



*Figure 6. Duplicate effect*

The reason the hypothesis failed is in the rows themselves. The removed duplicates are not copies of ordinary flows. Between 39% and 92% of them have an identical twin with a different label, against 2-10% of all rows, and their median flow has 0-2 packets (all flows: 8-43). They are near-empty flows that timing features can't tell apart. Keeping every copy multiplies errors that no model can avoid, which outweighs the advantage of having a test flow's copy in the training data:

Share of removed rows whose features also occur with another label, and packets per flow. A1 120 s packet counts use its published (corrupt) flowPktsPerSecond.

| task | timeout | duplicates removed | removed with other-label twin | all rows with other-label twin | median packets, removed | median packets, all |
|---|---|---|---|---|---|---|
| A1 | 15 | 720 | 0.761 | 0.032 | 0.0 | 32.0 |
| A1 | 30 | 1,041 | 0.51 | 0.039 | 2.0 | 22.0 |
| A1 | 60 | 838 | 0.766 | 0.047 | 0.0 | 16.0 |
| A1 | 120 | 1,257 | 0.434 | 0.056 | 0.0 | 0.0 |
| A2-VPN | 15 | 413 | 0.823 | 0.04 | 0.0 | 27.0 |
| A2-VPN | 30 | 790 | 0.395 | 0.045 | 2.0 | 17.0 |
| A2-VPN | 60 | 412 | 0.748 | 0.053 | 0.0 | 8.0 |
| A2-VPN | 120 | 983 | 0.45 | 0.098 | 2.0 | 8.0 |
| A2-NonVPN | 15 | 271 | 0.742 | 0.024 | 0.0 | 43.0 |
| A2-NonVPN | 30 | 223 | 0.897 | 0.03 | 0.0 | 26.0 |
| A2-NonVPN | 60 | 374 | 0.805 | 0.039 | 0.0 | 23.0 |
| A2-NonVPN | 120 | 210 | 0.919 | 0.039 | 0.0 | 17.0 |
| B | 15 | 684 | 0.804 | 0.035 | 0.0 | 32.0 |
| B | 30 | 1,013 | 0.529 | 0.042 | 2.0 | 22.0 |
| B | 60 | 786 | 0.818 | 0.053 | 0.0 | 16.0 |
| B | 120 | 1,193 | 0.562 | 0.078 | 2.0 | 12.0 |

The duplicate rule of data v1 stays: in our own split it guarantees that no flow is scored after being trained on, and this benchmark shows it changes 10-fold scores by only about +-0.02.

## 4.4 Storage after the baselines

Every official run saves its trained model (the plan's rule). After the first 147 runs (80 baselines, 64 benchmark, 2 pilot, 1 setup check), the MLflow store held 137.8 MB of artifacts and a 5.8 MB database (275.2 MB and 10.4 MB after the timing rerun of all 144 configs); the largest single run is 4.3 MB (SVM). Per model type:

| kind | runs | total_MB | per_run_max_MB | model_file_max_MB |
|---|---|---|---|---|
| knn (holdout) | 16 | 23.075 | 2.475 | 2.319 |
| knn_minmax (paper_10fold) | 32 | 66.742 | 3.596 | 3.438 |
| logreg (holdout) | 16 | 1.64 | 0.167 | 0.005 |
| majority (holdout) | 16 | 1.395 | 0.118 | 0.001 |
| svm_rbf (holdout) | 16 | 27.418 | 4.345 | 4.188 |
| tree (holdout) | 16 | 5.457 | 0.764 | 0.579 |
| tree (paper_10fold) | 32 | 11.674 | 0.817 | 0.629 |

Whole store now (after the rerun): artifacts 275.2 MB, mlflow.db 10.4 MB.

Tree ensembles in Phase 3 will be much larger: a 300-tree Random Forest, measured on the largest training set (B, 15 s), is a 171.9 MB file (22.4 MB with joblib compression level 3) [E03].

## 4.5 Phase 3: how the improvement experiments were run

Phase 3 (E04-E17) starts from the Phase 2 baselines and changes **one thing at a time** against the **current best** model of each task: the highest 15 s validation macro F1 so far, whatever the margin. Every experiment runs first at 15 s for all four tasks. A model is a **winner**, and is rerun at 30, 60 and 120 s with its 15 s settings, only if it beats the task's current best by at least 0.03; 0.03 is above the largest Phase 2 cross-validation standard deviation (0.025), so a smaller gain could be noise from the split. The rule was fixed on 28 Oct, before any Phase 3 run, and written into every experiment entry before it ran. All Phase 3 runs use data v1, the same dependency lock file and one thread per model; every model is saved with joblib compression level 3. Every table below is copied unchanged from `reports/phase3_tables.md`, generated by `scripts/results_tables.py --phase 3` from `reports/history.csv`; the figures come from notebooks 03 and 04 (`scripts/make_notebooks.py`). Scores are validation scores of the fixed 70/15/15 split (seed 42); the test set is not used in Phase 3.

## 4.6 E04: KNN with a signed log transform

**What.** KNN (k = 5, as in E03) with a signed log, sign(x)·log1p(|x|), applied inside the pipeline before scaling, once with StandardScaler and once with RobustScaler (Section 3.7). **Why.** KNN compares flows by distance, and a few very large timing values (up to about 1.2e8 microseconds) dominate every distance on the raw features. **Threshold (pre-stated).** (a) the better variant beats the E03 KNN by at least 0.02 in at least 3 of 4 tasks; (b) the two scalers are within 0.02 of each other in at least 3 of 4 tasks.

| task | model | val macro F1 | CV macro F1 | best before | best before F1 | gain | winner (gain >= 0.03) |
|---|---|---|---|---|---|---|---|
| A1 | E04-knn_log_robust | 0.8929 | 0.8874 +- 0.0042 | E02-tree | 0.8901 | 0.0028 | no |
| A1 | E04-knn_log_standard | 0.9048 | 0.9033 +- 0.0046 | E02-tree | 0.8901 | 0.0146 | no |
| A2-VPN | E04-knn_log_robust | 0.8372 | 0.8185 +- 0.0080 | E02-tree | 0.8247 | 0.0125 | no |
| A2-VPN | E04-knn_log_standard | 0.8544 | 0.8336 +- 0.0103 | E02-tree | 0.8247 | 0.0296 | no |
| A2-NonVPN | E04-knn_log_robust | 0.854 | 0.8688 +- 0.0050 | E02-tree | 0.8294 | 0.0247 | no |
| A2-NonVPN | E04-knn_log_standard | 0.867 | 0.8861 +- 0.0068 | E02-tree | 0.8294 | 0.0376 | yes |
| B | E04-knn_log_robust | 0.742 | 0.7555 +- 0.0094 | E02-tree | 0.7585 | -0.0166 | no |

| task | model | val macro F1 | CV macro F1 | best before | best before F1 | gain | winner (gain >= 0.03) |
|---|---|---|---|---|---|---|---|
| B | E04-knn_log_standard | 0.7644 | 0.7798 +- 0.0078 | E02-tree | 0.7585 | 0.0059 | no |

**Result.** The signed log lifts KNN by 0.080-0.104 in every task, and StandardScaler beats RobustScaler in every task (by 0.0118-0.0224). The signed-log KNN also passes the E02 tree in all four tasks and becomes the current best; only A2-NonVPN (+0.0376) is a winner (A2-VPN misses the margin by 0.0004). Rerun at the other timeouts, it stays the best A2-NonVPN run at each:

| task | timeout | model | val macro F1 | CV macro F1 | best other run | its F1 | difference |
|---|---|---|---|---|---|---|---|
| A2-NonVPN | 15 s | E04-knn_log_standard | 0.867 | 0.8861 +- 0.0068 | E04-knn_log_robust | 0.854 | 0.0129 |
| A2-NonVPN | 30 s | E04-knn_log_standard | 0.8523 | 0.8487 +- 0.0067 | E02-tree | 0.8283 | 0.024 |
| A2-NonVPN | 60 s | E04-knn_log_standard | 0.859 | 0.8638 +- 0.0108 | E02-tree | 0.8246 | 0.0344 |
| A2-NonVPN | 120 s | E04-knn_log_standard | 0.8491 | 0.8243 +- 0.0252 | E02-tree | 0.8393 | 0.0098 |

**Verdict.** (a) supported, (b) supported. [E04]

## 4.7 E05: Random Forest

**What.** A 300-tree Random Forest with default settings. **Why.** The E02 tree is unpruned (depth 16-35) and fits noise; averaging 300 trees grown on bootstrap samples lowers that variance. **Threshold.** (a) the forest beats the E02 tree by at least 0.03 in at least 3 of 4 tasks; (b) it beats the current best (the E04 KNN) in at least 2 of 4 tasks.

| task | model | val macro F1 | CV macro F1 | best before | best before F1 | gain | winner (gain >= 0.03) |
|---|---|---|---|---|---|---|---|
| A1 | E05-rf | 0.9203 | 0.9172 +- 0.0050 | E04-knn_log_standard | 0.9048 | 0.0156 | no |
| A2-VPN | E05-rf | 0.8679 | 0.8637 +- 0.0066 | E04-knn_log_standard | 0.8544 | 0.0135 | no |
| A2-NonVPN | E05-rf | 0.8941 | 0.9048 +- 0.0049 | E04-knn_log_standard | 0.867 | 0.0271 | no |
| B | E05-rf | 0.8147 | 0.8102 +- 0.0090 | E04-knn_log_standard | 0.7644 | 0.0503 | yes |

**Result.** The forest is the current best in all four tasks (+0.0135 to +0.0503 over the E04 KNN; +0.0302 to +0.0647 over the tree). Only B is a winner; its reruns:

| task | timeout | model | val macro F1 | CV macro F1 | best other run | its F1 | difference |
|---|---|---|---|---|---|---|---|
| B | 15 s | E05-rf | 0.8147 | 0.8102 +- 0.0090 | E04-knn_log_standard | 0.7644 | 0.0503 |
| B | 30 s | E05-rf | 0.8114 | 0.7979 +- 0.0119 | E02-tree | 0.7476 | 0.0638 |
| B | 60 s | E05-rf | 0.7771 | 0.7656 +- 0.0136 | E02-tree | 0.7058 | 0.0713 |
| B | 120 s | E05-rf | 0.7835 | 0.7562 +- 0.0083 | E02-tree | 0.7158 | 0.0677 |

**Verdict.** (a) and (b) supported. The forest is the largest model so far: 171.9 MiB uncompressed for B at 15 s, 22.4 MiB as a compressed file. [E05]

## 4.8 E06: gradient boosting (XGBoost and LightGBM)

**What.** XGBoost and LightGBM with default settings. **Why.** Boosted trees correct the errors of earlier trees, which should help the classes the forest still confuses. **Threshold.** (a) the better booster beats the forest in at least 2 of 4 tasks; (b) the two are within 0.02 of each other in at least 3 of 4 tasks.

| task | model | val macro F1 | CV macro F1 | best before | best before F1 | gain | winner (gain >= 0.03) |
|---|---|---|---|---|---|---|---|
| A1 | E06-lgbm | 0.9181 | 0.9175 +- 0.0042 | E05-rf | 0.9203 | -0.0022 | no |
| A1 | E06-xgb | 0.9271 | 0.9251 +- 0.0046 | E05-rf | 0.9203 | 0.0067 | no |
| A2-VPN | E06-lgbm | 0.8694 | 0.8827 +- 0.0080 | E05-rf | 0.8679 | 0.0015 | no |
| A2-VPN | E06-xgb | 0.8758 | 0.8799 +- 0.0025 | E05-rf | 0.8679 | 0.0079 | no |
| A2-NonVPN | E06-lgbm | 0.8983 | 0.9224 +- 0.0097 | E05-rf | 0.8941 | 0.0042 | no |
| A2-NonVPN | E06-xgb | 0.9015 | 0.9237 +- 0.0117 | E05-rf | 0.8941 | 0.0075 | no |
| B | E06-lgbm | 0.8201 | 0.8305 +- 0.0088 | E05-rf | 0.8147 | 0.0054 | no |
| B | E06-xgb | 0.8303 | 0.8334 +- 0.0053 | E05-rf | 0.8147 | 0.0156 | no |

**Result.** Default XGBoost is the current best in all four tasks (+0.0067 to +0.0156 over the forest); LightGBM is 0.0032-0.0102 below it. No winner. **Verdict.** (a) and (b) supported. Default XGBoost is also one of the cheapest models to run: 1.13-11.76 ms per 1,000 flows, against 21.91-27.28 ms for the forest. The figures show validation macro F1 with cross-validation mean +- std for every E02-E06 model at 15 s, the F1 of every class, and macro F1 against training and prediction time. [E06]

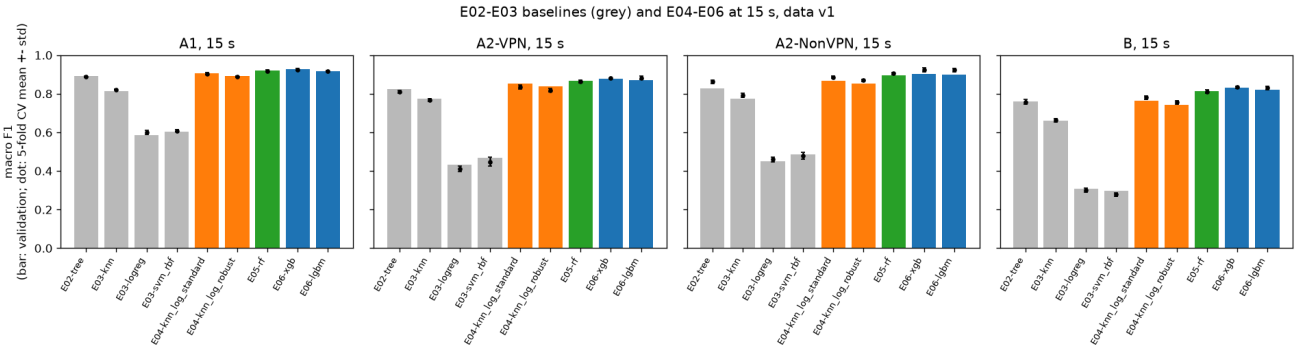



*Figure 7. E02-E06 at 15 s*

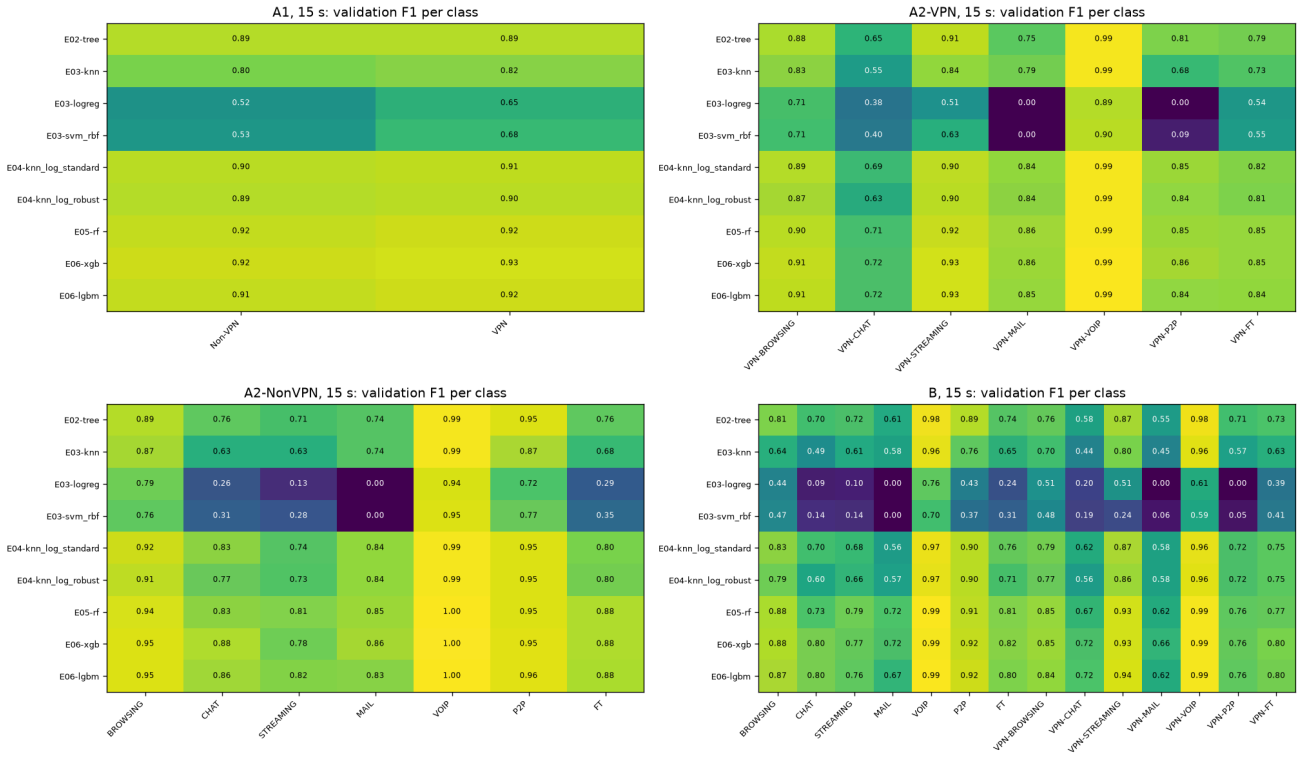



*Figure 8. Per-class F1, E02-E06*

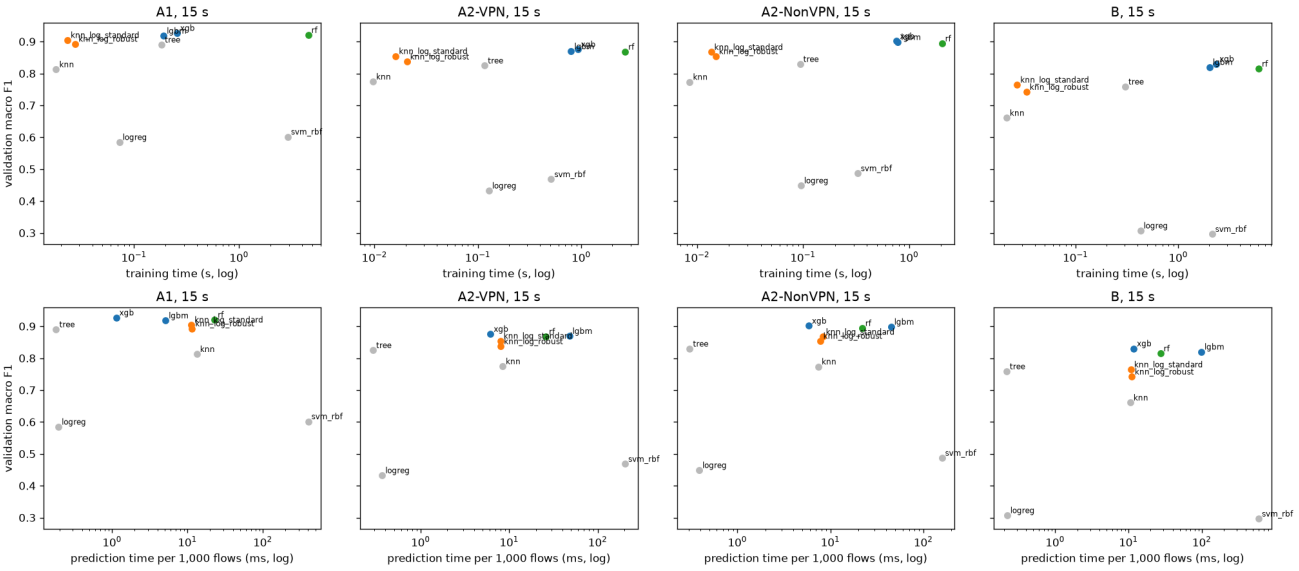



*Figure 9. Macro F1 against cost, E02-E06*

## 4.9 E07: hyperparameter tuning with Optuna

**What.** For each task the best 2 of Random Forest, XGBoost and LightGBM, plus the signed-log KNN, each tuned with the same budget: 50 Optuna trials (TPE sampler, seed 42), each trial scored by 5-fold cross-validation macro F1 on the training rows only. The search ranges were fixed in the plan before running. The best trial is retrained and scored on validation like any other run. **Why.** Defaults are general-purpose; giving every candidate the same budget means no model wins only because it was tuned more. **Threshold.** (a) tuned above untuned in at least 8 of 12 searches; (b) the best E07 model beats the current best (default XGBoost) by at least 0.01 in at least 2 of 4 tasks; (c) the tuned KNN stays below the best tuned tree ensemble in all four tasks.

| task | ranking (15 s val macro F1) | tuned in E07 | KNN preprocessing for E07 |
|---|---|---|---|
| A1 | xgb 0.9271, rf 0.9203, lgbm 0.9181 | xgb, rf | knn_log_standard 0.9048 |

| task | ranking (15 s val macro F1) | tuned in E07 | KNN preprocessing for E07 |
|---|---|---|---|
| A2-VPN | xgb 0.8758, lgbm 0.8694, rf 0.8679 | xgb, lgbm | knn_log_standard 0.8544 |
| A2-NonVPN | xgb 0.9015, lgbm 0.8983, rf 0.8941 | xgb, lgbm | knn_log_standard 0.8670 |
| B | xgb 0.8303, lgbm 0.8201, rf 0.8147 | xgb, lgbm | knn_log_standard 0.7644 |

| task | model | val macro F1 | CV macro F1 | best before | best before F1 | gain | winner (gain >= 0.03) |
|---|---|---|---|---|---|---|---|
| A1 | E07-knn_log_standard | 0.9163 | 0.9122 +- 0.0047 | E06-xgb | 0.9271 | -0.0108 | no |
| A1 | E07-rf | 0.9181 | 0.9209 +- 0.0025 | E06-xgb | 0.9271 | -0.009 | no |
| A1 | E07-xgb | 0.9259 | 0.9293 +- 0.0065 | E06-xgb | 0.9271 | -0.0011 | no |
| A2-VPN | E07-knn_log_standard | 0.8625 | 0.8575 +- 0.0048 | E06-xgb | 0.8758 | -0.0133 | no |
| A2-VPN | E07-lgbm | 0.8768 | 0.8902 +- 0.0023 | E06-xgb | 0.8758 | 0.001 | no |
| A2-VPN | E07-xgb | 0.8661 | 0.8838 +- 0.0011 | E06-xgb | 0.8758 | -0.0097 | no |
| A2-NonVPN | E07-knn_log_standard | 0.8929 | 0.9060 +- 0.0071 | E06-xgb | 0.9015 | -0.0086 | no |
| A2-NonVPN | E07-lgbm | 0.9028 | 0.9329 +- 0.0072 | E06-xgb | 0.9015 | 0.0013 | no |
| A2-NonVPN | E07-xgb | 0.9025 | 0.9294 +- 0.0067 | E06-xgb | 0.9015 | 0.001 | no |
| B | E07-knn_log_standard | 0.8006 | 0.8044 +- 0.0111 | E06-xgb | 0.8303 | -0.0297 | no |
| B | E07-lgbm | 0.8352 | 0.8447 +- 0.0058 | E06-xgb | 0.8303 | 0.0049 | no |
| B | E07-xgb | 0.8428 | 0.8405 +- 0.0056 | E06-xgb | 0.8303 | 0.0125 | no |

**Result.** Tuning raised the cross-validation score in all 12 searches but the validation score in only 9: the best of 50 cross-validation scores is optimistic. The largest gains are for KNN (B +0.0361, A2-NonVPN +0.0260); the ensembles gain at most +0.0151. Only B improves by more than 0.01 (tuned XGBoost 0.8428, +0.0125). No winner. In every search the retrained model's cross-validation score equals its best trial's exactly. **Verdict.** (a) supported (9 of 12), (b) not supported (1 of 4 tasks), (c) supported. [E07]

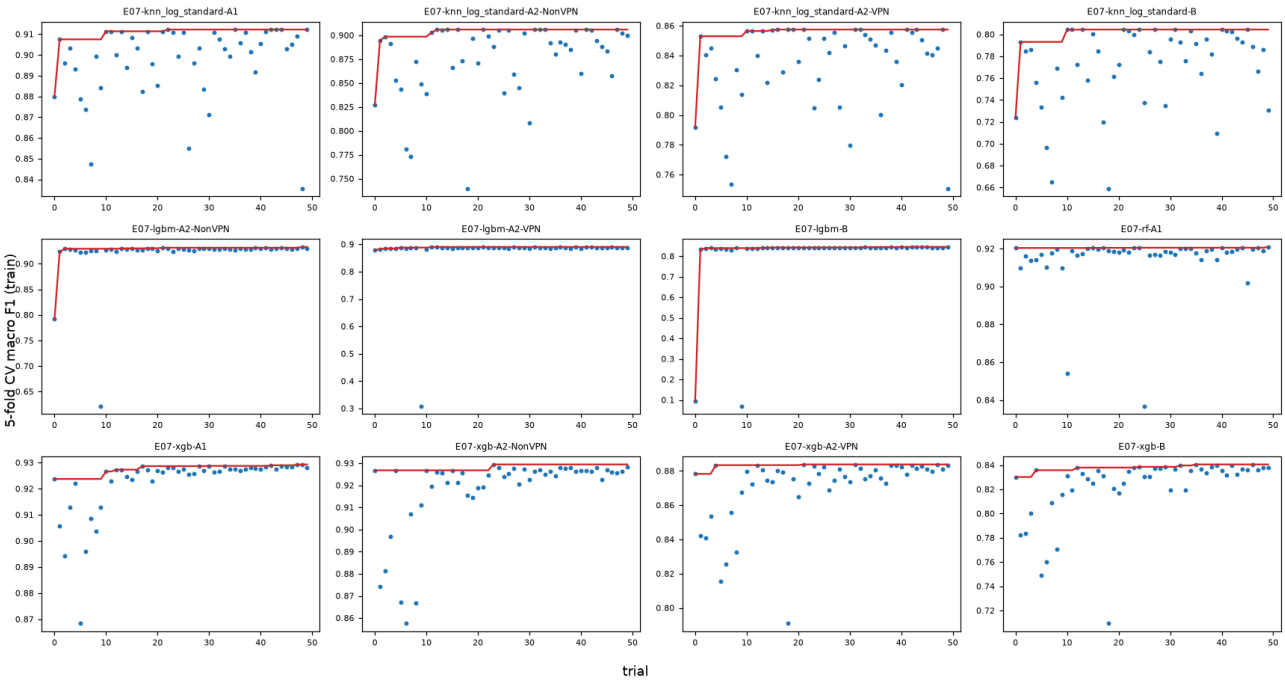



*Figure 10. E07 Optuna histories*

## 4.10 E08: class balancing

**What.** On the current best model per task, a ladder: none, then class weights, then random oversampling, then SMOTE (inside the pipeline, so only training folds are resampled); a step runs only if every earlier step helped. **Why.** Three tasks are imbalanced (largest to smallest class 5.79:1 in A2-VPN, 11.85:1 in A2-NonVPN and B), and macro F1 weights every class equally. **Threshold.** (a) the best step beats "none" by at least 0.01 in at least 2 of the 3 imbalanced tasks; (b) in A1 no step moves macro F1 by more than 0.01; (c) the best step raises the F1 of the weakest class in at least 2 of the 3 imbalanced tasks.

| task | step | run | val macro F1 | change vs none | F1 of weakest class without balancing | helped | next |
|---|---|---|---|---|---|---|---|
| A1 | none | E06-xgb | 0.9271 | 0.0 | Non-VPN 0.9240 | | |
| A1 | class_weight | E08-xgb_class_weight | 0.9237 | -0.0034 | Non-VPN 0.9202 | no | stop: this step did not help |
| A2-VPN | none | E07-lgbm | 0.8768 | 0.0 | VPN-CHAT 0.7343 | | |
| A2-VPN | class_weight | E08-lgbm_class_weight | 0.883 | 0.0062 | VPN-CHAT 0.7463 | yes | |
| A2-VPN | oversample | E08-lgbm_oversample | 0.883 | 0.0062 | VPN-CHAT 0.7477 | no | stop: this step did not help |
| A2-NonVPN | none | E07-lgbm | 0.9028 | 0.0 | STREAMING 0.8095 | | |
| A2-NonVPN | class_weight | E08-lgbm_class_weight | 0.9049 | 0.0022 | STREAMING 0.8189 | yes | |
| A2-NonVPN | oversample | E08-lgbm_oversample | 0.9075 | 0.0047 | STREAMING 0.8372 | yes | |
| A2-NonVPN | smote | E08-lgbm_smote | 0.9046 | 0.0019 | STREAMING 0.8217 | no | stop: this step did not help |
| B | none | E07-xgb | 0.8428 | 0.0 | VPN-MAIL 0.6496 | | |
| B | class_weight | E08-xgb_class_weight | 0.8364 | -0.0064 | VPN-MAIL 0.6290 | no | stop: this step did not help |

**Result.** Every change is within +-0.0064. Balancing helps a little in A2-VPN (class weights +0.0062) and A2-NonVPN (random oversampling +0.0047) and hurts in A1 and B. **Verdict.** (a) not supported, (b) supported, (c)

supported. The weak classes (VPN-CHAT, STREAMING, VPN-MAIL) seem to fail because their flows look like those of other classes, not because there are too few of them; E08 does not test this directly. [E08]

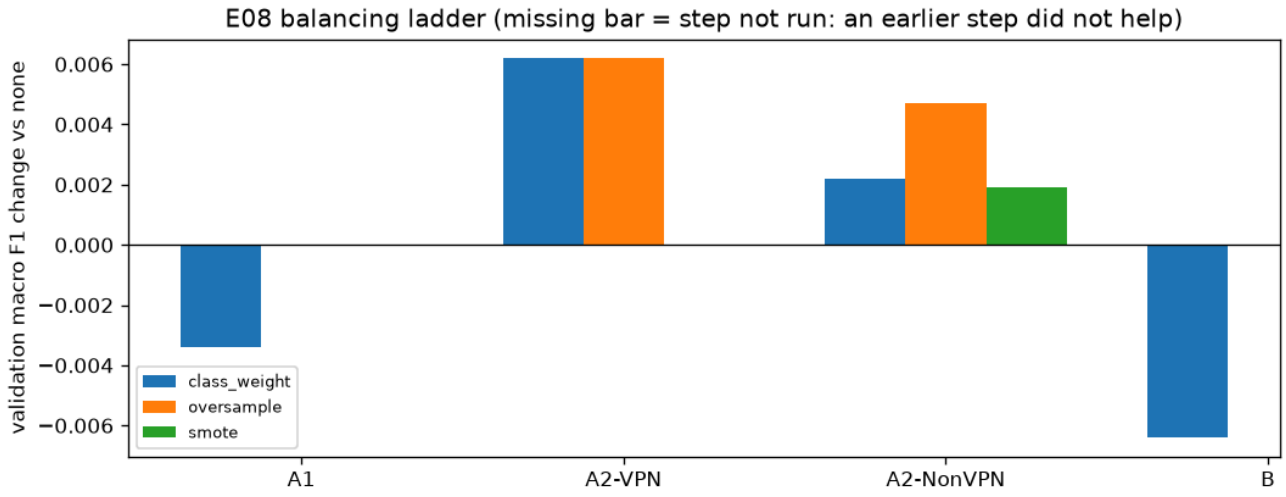



*Figure 11. E08 balancing ladder*

## 4.11 E10: task B as one 14-class model or two steps

**What.** A two-step model for task B: a VPN / non-VPN model routes each flow to one of two 7-class models. All three models use the current best B model type and settings (tuned XGBoost), and all are trained on B's training rows only. They are never trained on the A1 split, because A1 and B hold the same flows split differently. **Why.** Task B mixes two questions; separating them could help. **Threshold.** (a) the two-step score is within 0.01 of the direct model; (b) its first step makes fewer VPN / non-VPN errors than the direct model (180); (c) more than half of its errors come from a wrong first step.

| structure | run | val macro F1 | CV macro F1 | val accuracy | rows | errors | group errors (VPN / non-VPN wrong) | type errors (right group, wrong type) | group accuracy | share of errors from the group |
|---|---|---|---|---|---|---|---|---|---|---|
| direct 14-class | E07-xgb | 0.8428 | 0.8405 +- 0.0056 | 0.8794 | 2,711 | 327 | 180 | 147 | 0.9336 | 0.5505 |
| two-step | E10-xgb_two_step | 0.8318 | 0.8274 +- 0.0083 | 0.8727 | 2,711 | 345 | 191 | 154 | 0.9295 | 0.5536 |

**Result.** The two-step model scores 0.0110 lower, makes more group errors (191 vs 180), and 55.36% of its errors come from the first step. The direct model also makes 55.05% of its errors across the VPN / non-VPN line: telling VPN from non-VPN is the hard part either way. **Verdict.** (a) not supported, (b) not supported, (c) supported. Task B stays a direct 14-class model. [E10]

## 4.12 E17: the same tuning budget for every method type

**What.** The same 50-trial search as E07 for every method type not yet tuned: decision tree, logistic regression and SVM-RBF in all four tasks, Random Forest in A2-VPN, A2-NonVPN and B, and LightGBM in A1 (16 searches). Logistic regression and SVM may choose the signed log inside the search. **Why.** Without it, a comparison of method types would mix tuned and untuned models, the weakness we point out in earlier studies. Added on 28 Oct after E10, with the ranges agreed before running. **Threshold.** (a) tuned above untuned in at least 12 of 16 searches; (b) the best run per task is still XGBoost or LightGBM in at least 3 of 4 tasks; (c) the signed log is chosen in at least 6 of the 8 logistic-regression and SVM searches.

| task | model | val macro F1 | CV macro F1 | best before | best before F1 | gain | winner (gain >= 0.03) |
|---|---|---|---|---|---|---|---|
| A1 | E17-lgbm | 0.9308 | 0.9320 +- 0.0028 | E06-xgb | 0.9271 | 0.0037 | no |

| task | model | val macro F1 | CV macro F1 | best before | best before F1 | gain | winner (gain >= 0.03) |
|---|---|---|---|---|---|---|---|
| A1 | E17-logreg | 0.6517 | 0.6762 +- 0.0051 | E06-xgb | 0.9271 | -0.2754 | no |
| A1 | E17-svm_rbf | 0.9259 | 0.9187 +- 0.0044 | E06-xgb | 0.9271 | -0.0011 | no |
| A1 | E17-tree | 0.8905 | 0.8915 +- 0.0043 | E06-xgb | 0.9271 | -0.0366 | no |
| A2-VPN | E17-logreg | 0.7263 | 0.7344 +- 0.0074 | E08-lgbm_class_weight | 0.883 | -0.1566 | no |
| A2-VPN | E17-rf | 0.8704 | 0.8707 +- 0.0069 | E08-lgbm_class_weight | 0.883 | -0.0126 | no |
| A2-VPN | E17-svm_rbf | 0.8707 | 0.8668 +- 0.0074 | E08-lgbm_class_weight | 0.883 | -0.0123 | no |
| A2-VPN | E17-tree | 0.8292 | 0.8129 +- 0.0071 | E08-lgbm_class_weight | 0.883 | -0.0538 | no |
| A2-NonVPN | E17-logreg | 0.8002 | 0.8138 +- 0.0120 | E08-lgbm_oversample | 0.9075 | -0.1073 | no |
| A2-NonVPN | E17-rf | 0.8829 | 0.9063 +- 0.0050 | E08-lgbm_oversample | 0.9075 | -0.0246 | no |
| A2-NonVPN | E17-svm_rbf | 0.9058 | 0.9256 +- 0.0042 | E08-lgbm_oversample | 0.9075 | -0.0017 | no |
| A2-NonVPN | E17-tree | 0.8294 | 0.8619 +- 0.0086 | E08-lgbm_oversample | 0.9075 | -0.0781 | no |
| B | E17-logreg | 0.5862 | 0.5909 +- 0.0067 | E07-xgb | 0.8428 | -0.2566 | no |
| B | E17-rf | 0.8167 | 0.8270 +- 0.0092 | E07-xgb | 0.8428 | -0.0261 | no |
| B | E17-svm_rbf | 0.8242 | 0.8228 +- 0.0059 | E07-xgb | 0.8428 | -0.0186 | no |
| B | E17-tree | 0.7585 | 0.7575 +- 0.0132 | E07-xgb | 0.8428 | -0.0843 | no |

**Result.** SVM-RBF gains the most (+0.3243 to +0.5277), and every logistic-regression and SVM search chose the signed log. The tuned SVM ends 0.0017-0.0186 below each task's best. Logistic regression gains +0.0670 to +0.3515 but stays at 0.5862-0.8002. The tree (0.0000 to +0.0044) and the forest (-0.0111 to +0.0025) barely change. Tuned LightGBM is the new A1 best (0.9308). No winner. **Verdict.** (a) supported (13 of 16), (b) supported (4 of 4), (c) supported (8 of 8). The A2-NonVPN logistic regression stops at its fixed 1,000 iterations before converging (chosen C = 985); this is recorded as a limitation and was not changed after the run. [E17]

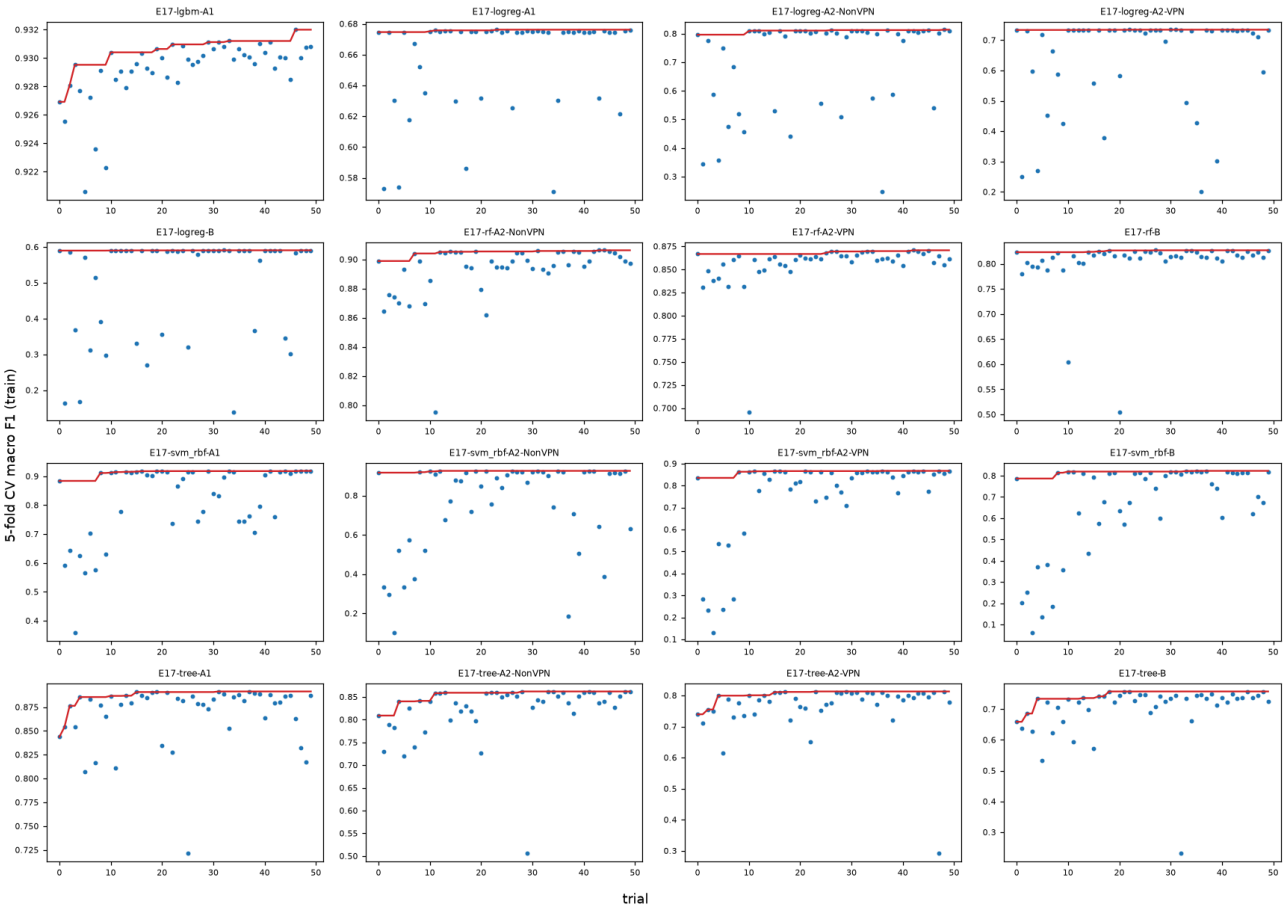



*Figure 12. E17 Optuna histories*

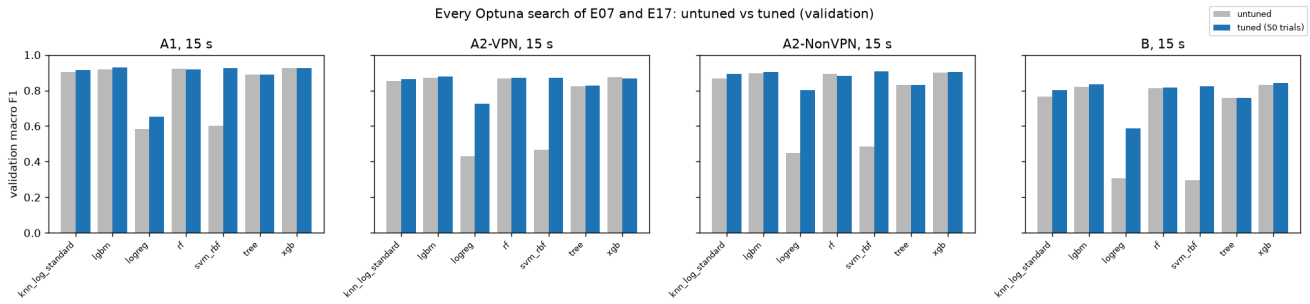



*Figure 13. Untuned vs tuned, E07 and E17*

## 4.13 Final choice per task and the comparison set

The final configuration per task is the run with the highest validation macro F1 among all model-selection runs (E02-E17, any timeout), decided on 28 Oct after E10. This replaces the rule set before Phase 3, under which any gap below 0.03 was a tie broken in favour of the simpler model. The cross-validation standard deviations of the headline models (0.0028-0.0058) are far below 0.03, and under the old rule KNN or the SVM would have been chosen at 0.0146-0.0286 below the best. This is a deviation from the pre-set rule and is reported as one (Section 3.8). The models the old rule would pick are also scored on the test set in Phase 4 (E12), so the cost of either rule is visible. The only fallback is a saved model file above 50 MB; none of the four is.

| task | run | method type | timeout | val macro F1 | next best | next best F1 | model file MiB | skipped for file size |
|---|---|---|---|---|---|---|---|---|
| A1 | E17-lgbm-A1-15s | LightGBM | 15 | 0.9308 | E06-xgb-A1-15s | 0.9271 | 7.09 | none |
| A2-VPN | E08-lgbm_class_weight-A2-VPN-15s | LightGBM | 15 | 0.883 | E08-lgbm_oversample-A2-VPN-15s | 0.883 | 25.12 | none |

| task | run | method type | timeout | val macro F1 | next best | next best F1 | model file MiB | skipped for file size |
|---|---|---|---|---|---|---|---|---|
| A2-NonVPN | E08-lgbm_oversample-A2-NonVPN-15s | LightGBM | 15 | 0.9075 | E17-svm_rbf-A2-NonVPN-15s | 0.9058 | 4.32 | none |
| B | E07-xgb-B-15s | XGBoost | 15 | 0.8428 | E08-xgb_class_weight-B-15s | 0.8364 | 4.31 | none |

| task | run | method type | val macro F1 | below the best by | runs within 0.03 | in comparison set |
|---|---|---|---|---|---|---|
| A1 | E04-knn_log_standard-A1-15s | KNN | 0.9048 | 0.026 | 10 | no |
| A2-VPN | E04-knn_log_standard-A2-VPN-15s | KNN | 0.8544 | 0.0286 | 11 | no |
| A2-NonVPN | E07-knn_log_standard-A2-NonVPN-15s | KNN | 0.8929 | 0.0146 | 11 | yes |
| B | E17-svm_rbf-B-15s | SVM-RBF | 0.8242 | 0.0186 | 9 | yes |

The **comparison set** holds, per task, the best 15 s validation variant of each of the 7 method types, 28 models in all. It was fixed in `configs/final/compare/` before the test set is opened. In Phase 4 each is retrained on training + validation data and scored once on the test set, with paired bootstrap tests (Holm correction) for differences.

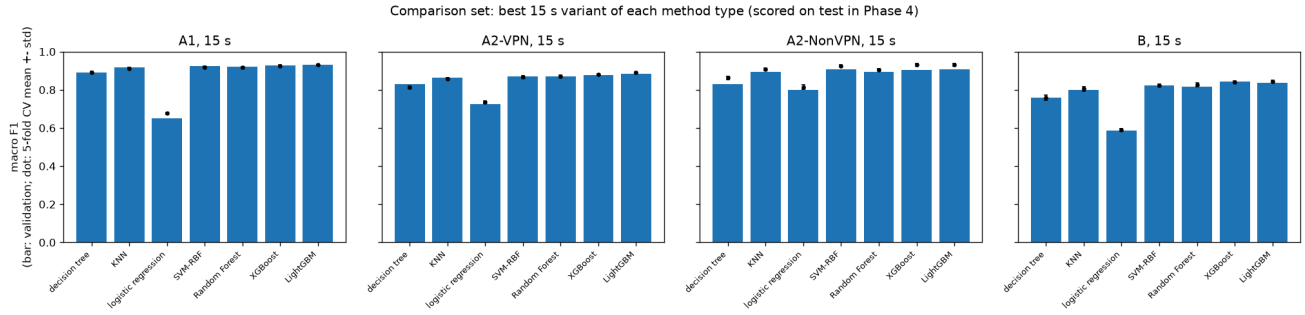



*Figure 14. Comparison set at 15 s*

| task | method type | run | val macro F1 | variants compared |
|---|---|---|---|---|
| A1 | decision tree | E17-tree-A1-15s | 0.8905 | 2 |
| A1 | KNN | E07-knn_log_standard-A1-15s | 0.9163 | 4 |
| A1 | logistic regression | E17-logreg-A1-15s | 0.6517 | 2 |
| A1 | SVM-RBF | E17-svm_rbf-A1-15s | 0.9259 | 2 |
| A1 | Random Forest | E05-rf-A1-15s | 0.9203 | 2 |
| A1 | XGBoost | E06-xgb-A1-15s | 0.9271 | 3 |
| A1 | LightGBM | E17-lgbm-A1-15s | 0.9308 | 2 |
| A2-VPN | decision tree | E17-tree-A2-VPN-15s | 0.8292 | 2 |
| A2-VPN | KNN | E07-knn_log_standard-A2-VPN-15s | 0.8625 | 4 |
| A2-VPN | logistic regression | E17-logreg-A2-VPN-15s | 0.7263 | 2 |
| A2-VPN | SVM-RBF | E17-svm_rbf-A2-VPN-15s | 0.8707 | 2 |
| A2-VPN | Random Forest | E17-rf-A2-VPN-15s | 0.8704 | 2 |
| A2-VPN | XGBoost | E06-xgb-A2-VPN-15s | 0.8758 | 2 |
| A2-VPN | LightGBM | E08-lgbm_class_weight-A2-VPN-15s | 0.883 | 4 |
| A2-NonVPN | decision tree | E02-tree-A2-NonVPN-15s | 0.8294 | 2 |
| A2-NonVPN | KNN | E07-knn_log_standard-A2-NonVPN-15s | 0.8929 | 4 |

| task | method type | run | val macro F1 | variants compared |
|---|---|---|---|---|
| A2-NonVPN | logistic regression | E17-logreg-A2-NonVPN-15s | 0.8002 | 2 |
| A2-NonVPN | SVM-RBF | E17-svm_rbf-A2-NonVPN-15s | 0.9058 | 2 |
| A2-NonVPN | Random Forest | E05-rf-A2-NonVPN-15s | 0.8941 | 2 |
| A2-NonVPN | XGBoost | E07-xgb-A2-NonVPN-15s | 0.9025 | 2 |
| A2-NonVPN | LightGBM | E08-lgbm_oversample-A2-NonVPN-15s | 0.9075 | 5 |
| B | decision tree | E02-tree-B-15s | 0.7585 | 2 |
| B | KNN | E07-knn_log_standard-B-15s | 0.8006 | 4 |
| B | logistic regression | E17-logreg-B-15s | 0.5862 | 2 |
| B | SVM-RBF | E17-svm_rbf-B-15s | 0.8242 | 2 |
| B | Random Forest | E17-rf-B-15s | 0.8167 | 2 |
| B | XGBoost | E07-xgb-B-15s | 0.8428 | 4 |
| B | LightGBM | E07-lgbm-B-15s | 0.8352 | 2 |

## 4.14 E09: feature-group ablation (analysis only)

**What.** The four headline models retrained with one feature group only, with one group removed, and with the timing-only set (all but flowBytesPerSecond, the only size-based feature). **Threshold.** (a) every group alone scores below all 23 features in all four tasks; (b) dropping any one group costs at most 0.02 in at least 3 of 4 tasks; (c) the timing-only set is within 0.01 of all 23 features in at least 3 of 4 tasks. If any set scored at least 0.03 above all 23, the work would stop and the user would be asked (it did not happen).

| task | feature set | features | val macro F1 | change vs all | CV macro F1 |
|---|---|---|---|---|---|
| A1 | all | 23 | 0.9308 | 0.0 | 0.9320 +- 0.0028 |
| A1 | only_fiat | 4 | 0.8078 | -0.1229 | 0.8111 +- 0.0059 |
| A1 | only_biat | 4 | 0.8243 | -0.1064 | 0.8136 +- 0.0100 |
| A1 | only_flowiat | 4 | 0.8507 | -0.08 | 0.8456 +- 0.0043 |
| A1 | only_active | 4 | 0.5489 | -0.3819 | 0.5618 +- 0.0124 |
| A1 | only_idle | 4 | 0.5505 | -0.3803 | 0.5614 +- 0.0107 |
| A1 | only_rates_duration | 3 | 0.8392 | -0.0915 | 0.8363 +- 0.0063 |
| A1 | no_fiat | 19 | 0.9256 | -0.0052 | 0.9187 +- 0.0050 |
| A1 | no_biat | 19 | 0.9152 | -0.0155 | 0.9161 +- 0.0050 |
| A1 | no_flowiat | 19 | 0.9322 | 0.0014 | 0.9222 +- 0.0019 |
| A1 | no_active | 19 | 0.9296 | -0.0011 | 0.9300 +- 0.0024 |
| A1 | no_idle | 19 | 0.9304 | -0.0004 | 0.9302 +- 0.0031 |
| A1 | no_rates_duration | 20 | 0.9185 | -0.0122 | 0.9133 +- 0.0054 |
| A1 | timing_only | 22 | 0.9159 | -0.0148 | 0.9140 +- 0.0031 |
| A2-VPN | all | 23 | 0.883 | 0.0 | 0.8893 +- 0.0044 |
| A2-VPN | only_fiat | 4 | 0.7593 | -0.1237 | 0.7326 +- 0.0146 |
| A2-VPN | only_biat | 4 | 0.7199 | -0.1631 | 0.6958 +- 0.0108 |
| A2-VPN | only_flowiat | 4 | 0.7861 | -0.0969 | 0.7690 +- 0.0107 |
| A2-VPN | only_active | 4 | 0.3252 | -0.5578 | 0.3222 +- 0.0193 |
| A2-VPN | only_idle | 4 | 0.3067 | -0.5763 | 0.3208 +- 0.0145 |
| A2-VPN | only_rates_duration | 3 | 0.779 | -0.104 | 0.7661 +- 0.0082 |
| A2-VPN | no_fiat | 19 | 0.8651 | -0.0179 | 0.8723 +- 0.0037 |

| task | feature set | features | val macro F1 | change vs all | CV macro F1 |
|---|---|---|---|---|---|
| A2-VPN | no_biat | 19 | 0.8724 | -0.0105 | 0.8760 +- 0.0037 |
| A2-VPN | no_flowiat | 19 | 0.8752 | -0.0078 | 0.8833 +- 0.0054 |
| A2-VPN | no_active | 19 | 0.886 | 0.003 | 0.8895 +- 0.0041 |
| A2-VPN | no_idle | 19 | 0.8826 | -0.0004 | 0.8893 +- 0.0054 |
| A2-VPN | no_rates_duration | 20 | 0.8435 | -0.0395 | 0.8449 +- 0.0083 |
| A2-VPN | timing_only | 22 | 0.849 | -0.034 | 0.8410 +- 0.0095 |
| A2-NonVPN | all | 23 | 0.9075 | 0.0 | 0.9310 +- 0.0058 |
| A2-NonVPN | only_fiat | 4 | 0.7535 | -0.154 | 0.7720 +- 0.0087 |
| A2-NonVPN | only_biat | 4 | 0.7271 | -0.1804 | 0.7407 +- 0.0140 |
| A2-NonVPN | only_flowiat | 4 | 0.8165 | -0.091 | 0.8288 +- 0.0095 |
| A2-NonVPN | only_active | 4 | 0.3355 | -0.572 | 0.3208 +- 0.0128 |
| A2-NonVPN | only_idle | 4 | 0.3038 | -0.6037 | 0.3225 +- 0.0177 |
| A2-NonVPN | only_rates_duration | 3 | 0.8471 | -0.0604 | 0.8396 +- 0.0110 |
| A2-NonVPN | no_fiat | 19 | 0.9024 | -0.0051 | 0.9260 +- 0.0087 |
| A2-NonVPN | no_biat | 19 | 0.8993 | -0.0082 | 0.9240 +- 0.0087 |
| A2-NonVPN | no_flowiat | 19 | 0.903 | -0.0045 | 0.9189 +- 0.0065 |
| A2-NonVPN | no_active | 19 | 0.9093 | 0.0018 | 0.9283 +- 0.0066 |
| A2-NonVPN | no_idle | 19 | 0.9039 | -0.0036 | 0.9300 +- 0.0045 |
| A2-NonVPN | no_rates_duration | 20 | 0.8774 | -0.0301 | 0.8980 +- 0.0037 |
| A2-NonVPN | timing_only | 22 | 0.8744 | -0.0331 | 0.9000 +- 0.0060 |
| B | all | 23 | 0.8428 | 0.0 | 0.8405 +- 0.0056 |
| B | only_fiat | 4 | 0.64 | -0.2028 | 0.6399 +- 0.0089 |
| B | only_biat | 4 | 0.6098 | -0.233 | 0.6147 +- 0.0121 |
| B | only_flowiat | 4 | 0.6716 | -0.1712 | 0.6926 +- 0.0102 |
| B | only_active | 4 | 0.2138 | -0.629 | 0.2262 +- 0.0072 |
| B | only_idle | 4 | 0.2083 | -0.6345 | 0.2118 +- 0.0092 |
| B | only_rates_duration | 3 | 0.7043 | -0.1385 | 0.6886 +- 0.0091 |
| B | no_fiat | 19 | 0.8244 | -0.0184 | 0.8266 +- 0.0072 |
| B | no_biat | 19 | 0.8163 | -0.0265 | 0.8189 +- 0.0067 |
| B | no_flowiat | 19 | 0.8262 | -0.0166 | 0.8284 +- 0.0052 |
| B | no_active | 19 | 0.8479 | 0.0051 | 0.8400 +- 0.0071 |
| B | no_idle | 19 | 0.8428 | 0.0 | 0.8405 +- 0.0058 |
| B | no_rates_duration | 20 | 0.7879 | -0.0549 | 0.7936 +- 0.0076 |
| B | timing_only | 22 | 0.7987 | -0.0441 | 0.7949 +- 0.0062 |

**Result.** No single group comes close to all 23 features (the best single group is 0.0604-0.1385 below). Rates + duration matter most: removing them costs 0.0122-0.0549. Active or idle features can be removed with almost no effect (at most +-0.0051). The timing-only set loses 0.0148-0.0441, so packet sizes carry information here.
**Verdict.** (a) supported, (b) not supported, (c) not supported. Nothing changes in the final configurations. [E09]

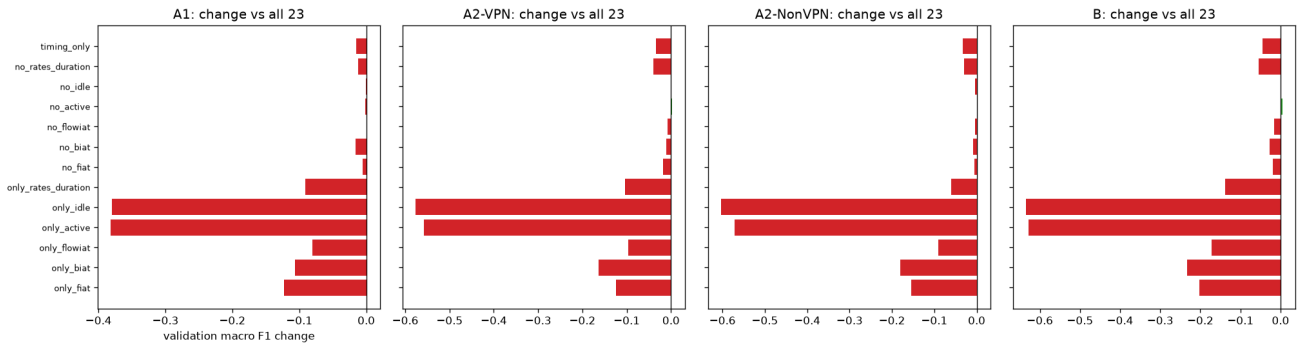



*Figure 15. E09 feature-group ablation*

## 4.15 E13: learning curve (analysis only)

**What.** The headline models trained on 10%, 20%, 40%, 60% and 80% of the training rows (stratified samples, 5 seeds each) and scored on the full validation set; 100% is the headline run. **Threshold.** (a) 100% scores above 10% in all four tasks; (b) the gain from 80% to 100% is below 0.01 in at least 3 of 4 tasks; (c) the spread over seeds is larger at 10% than at 80% in at least 3 of 4 tasks.

| task | train fraction | runs | training rows | val macro F1 mean | val macro F1 std |
|---|---|---|---|---|---|
| A1 | 0.1 | 5 | 1,262 | 0.8724 | 0.0062 |
| A1 | 0.2 | 5 | 2,525 | 0.8952 | 0.0041 |
| A1 | 0.4 | 5 | 5,050 | 0.9106 | 0.0066 |
| A1 | 0.6 | 5 | 7,575 | 0.9213 | 0.0039 |
| A1 | 0.8 | 5 | 10,100 | 0.927 | 0.0029 |
| A1 | 1.0 | 1 | 12,626 | 0.9308 | |
| A2-VPN | 0.1 | 5 | 656 | 0.7793 | 0.0052 |
| A2-VPN | 0.2 | 5 | 1,313 | 0.826 | 0.0101 |
| A2-VPN | 0.4 | 5 | 2,626 | 0.8582 | 0.0041 |
| A2-VPN | 0.6 | 5 | 3,939 | 0.8721 | 0.0053 |
| A2-VPN | 0.8 | 5 | 5,252 | 0.8804 | 0.0036 |
| A2-VPN | 1.0 | 1 | 6,566 | 0.883 | |
| A2-NonVPN | 0.1 | 5 | 608 | 0.8176 | 0.0093 |
| A2-NonVPN | 0.2 | 5 | 1,217 | 0.8585 | 0.0118 |
| A2-NonVPN | 0.4 | 5 | 2,434 | 0.8806 | 0.0057 |
| A2-NonVPN | 0.6 | 5 | 3,651 | 0.8934 | 0.0045 |
| A2-NonVPN | 0.8 | 5 | 4,868 | 0.9005 | 0.0033 |
| A2-NonVPN | 1.0 | 1 | 6,085 | 0.9075 | |
| B | 0.1 | 5 | 1,265 | 0.6969 | 0.0097 |
| B | 0.2 | 5 | 2,530 | 0.7468 | 0.0138 |
| B | 0.4 | 5 | 5,060 | 0.7975 | 0.0036 |
| B | 0.6 | 5 | 7,590 | 0.8156 | 0.0016 |
| B | 0.8 | 5 | 10,120 | 0.8297 | 0.0029 |
| B | 1.0 | 1 | 12,651 | 0.8428 | |

**Result.** Every task gains with more data (10% to 100%: +0.0584 to +0.1459). A1 and both A2 tasks are near their plateau (80% to 100%: +0.0026 to +0.0070), but the 14-class task B still gains +0.0131. **Verdict.** (a), (b) and (c) supported. [E13]

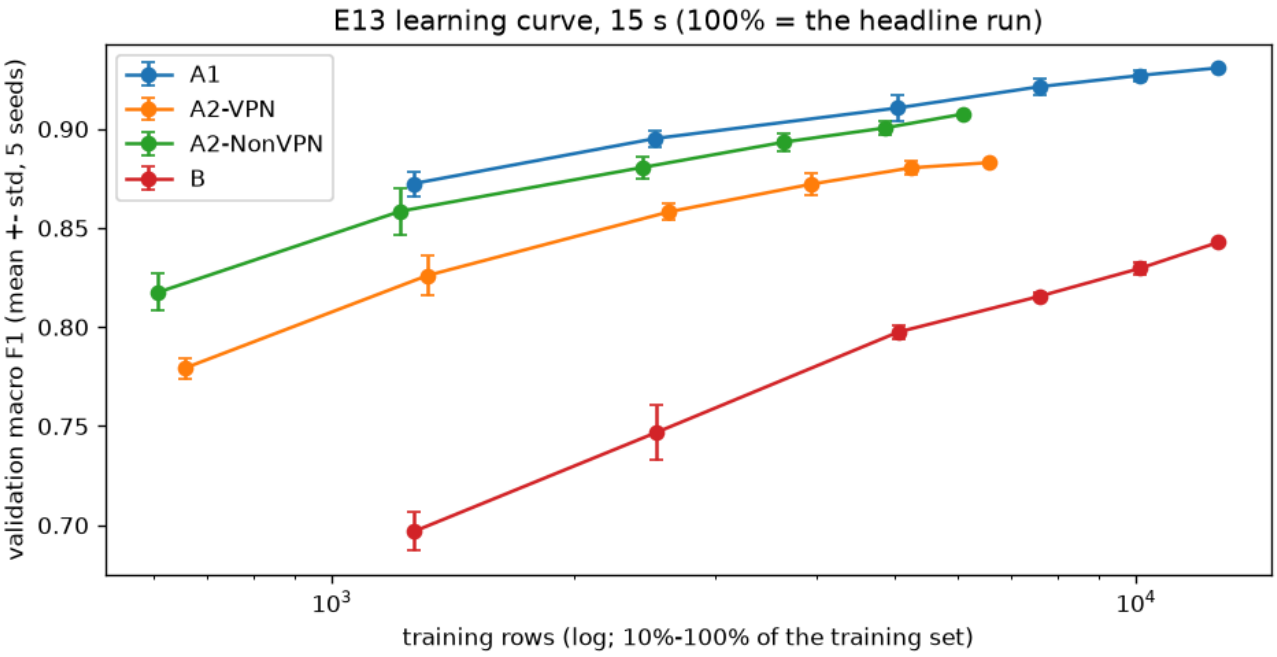



*Figure 16. E13 learning curve*

## 4.16 E14: split-seed stability (analysis only)

**What.** The headline models retrained on 10 new stratified splits of training + validation (seeds 42-51, same sizes). The seed-42 test rows are never pooled or looked at. Seed 42 here is a resplit, not the official split. **Threshold.** (a) the standard deviation is below 0.01 in at least 3 of 4 tasks; (b) mean cross-validation minus mean validation is within 0.01 in at least 3 of 4 tasks; (c) the official validation score lies inside the range of the 10 resplits in all four tasks.

| task | seeds | val macro F1 mean | val macro F1 std | val macro F1 min | val macro F1 max | val accuracy mean | val accuracy std | CV macro F1 mean | CV - val mean | official split (seed 42) val macro F1 |
|---|---|---|---|---|---|---|---|---|---|---|
| A1 | 10 | 0.9338 | 0.0042 | 0.9275 | 0.9397 | 0.9339 | 0.0042 | 0.9306 | -0.0032 | 0.9308 |
| A2-VPN | 10 | 0.8862 | 0.0114 | 0.8689 | 0.902 | 0.9037 | 0.0066 | 0.8805 | -0.0057 | 0.883 |
| A2-NonVPN | 10 | 0.9324 | 0.0071 | 0.9257 | 0.9468 | 0.9589 | 0.0044 | 0.9283 | -0.0041 | 0.9075 |
| B | 10 | 0.8476 | 0.0084 | 0.8344 | 0.8636 | 0.8819 | 0.0074 | 0.8358 | -0.0119 | 0.8428 |

**Result.** Scores move with the split by a standard deviation of 0.0042-0.0114. In A2-NonVPN the official validation score (0.9075) is below all 10 resplits (0.9257-0.9468): its validation rows are unusually hard, mostly for MAIL and P2P. This also explains the large cross-validation minus validation gaps seen for A2-NonVPN in E04-E17. **Verdict.** (a) supported, (b) supported, (c) not supported. [E14]

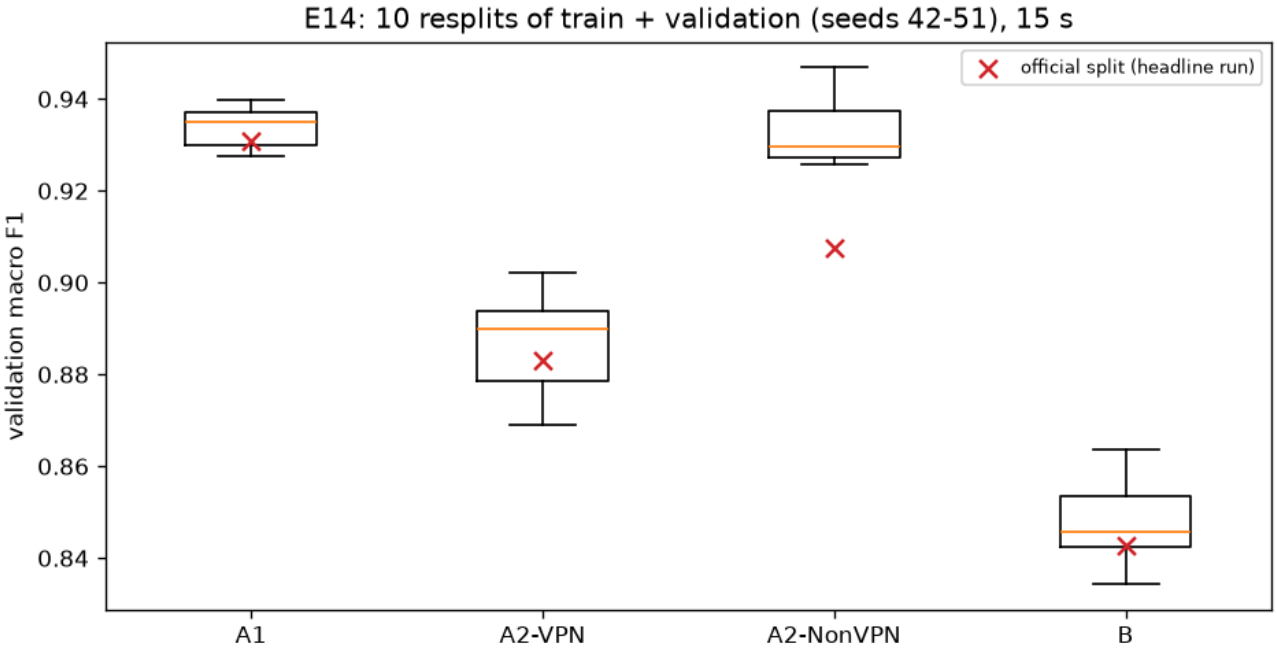



*Figure 17. E14 seed stability*

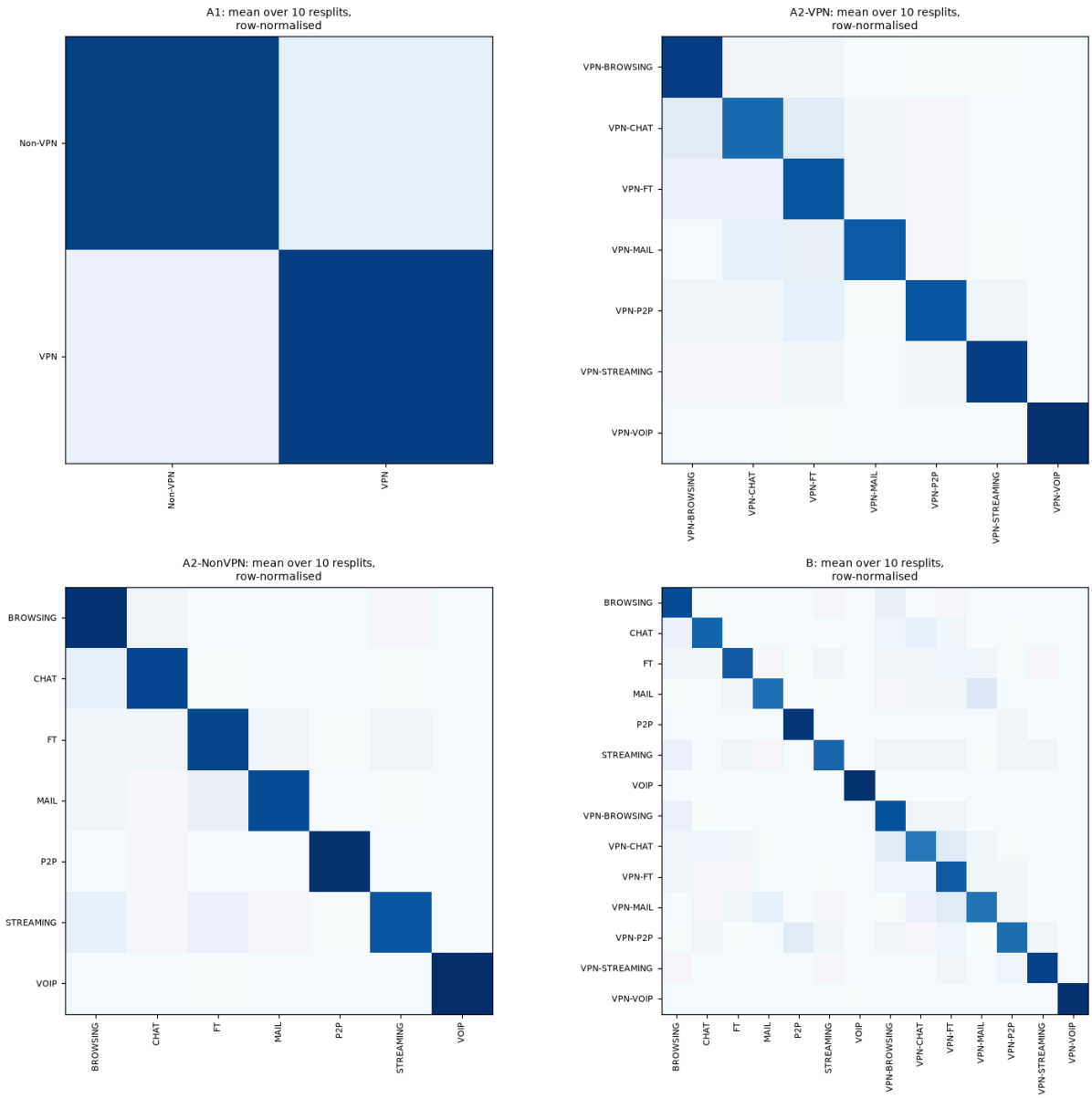



*Figure 18. E14 mean confusion matrices*

## 4.17 E11: session-aware split (not run)

The ARFF files hold only the 23 features and the label, with no flow IDs, addresses, ports, timestamps or capture-file names, so flows cannot be grouped by recording session (E00). E11 therefore does not run. Exact duplicates were removed before the split (E00), and E14 and E16(b) measure related effects. The missing session split is a limitation (Section 7). [E11]

## 4.18 E16: reproducing Isnan et al. (2026), then correcting it

**What.** Isnan et al. report 88.01% accuracy for a 300-tree Random Forest on the Kaggle copy of the dataset, which is the eight A2 files from all four timeouts stacked (Section 2.9). E16 used its own data version (kaggle-k1) and ran four parts:

- **(a)** their exact setup: duplicates dropped over the whole file, a stratified 80/20 split with seed 42, and their grid search with accuracy as the score;
- **(b)** the same method with each timeout kept apart;
- **(c)** our tuned task-B XGBoost under (b);
- **(d)**, added in a recheck: (a)'s forest trained on mixed-timeout rows cut to each per-timeout training size.

**Threshold.** (a) accuracy and weighted F1 within 0.01 of theirs; (b) the per-timeout mean is at least 0.01 below (a); (c) our model's per-timeout mean is at least the forest's; (d) at equal training size, mixing timeouts adds less than 0.01 over (b).

| run | split | train / test rows | grid pick | grid best CV accuracy | test accuracy | test weighted F1 | test macro F1 |
|---|---|---|---|---|---|---|---|
| E16-isnan-rf | isnan | 32,789 / 8,198 | {"max_depth": null, "min_samples_leaf": 1, "min_samples_split": 2, "n_estimators": 200} | 0.8742 | 0.8796 | 0.8792 | 0.8407 |
| E16-per_timeout-rf-120s | per_timeout_120s | 7,671 / 1,918 | {"max_depth": null, "min_samples_leaf": 1, "min_samples_split": 5, "n_estimators": 300} | 0.8178 | 0.8206 | 0.817 | 0.7724 |
| E16-per_timeout-rf-15s | per_timeout_15s | 14,459 / 3,615 | {"max_depth": 30, "min_samples_leaf": 1, "min_samples_split": 2, "n_estimators": 300} | 0.8578 | 0.8658 | 0.8652 | 0.8323 |
| E16-per_timeout-rf-30s | per_timeout_30s | 10,910 / 2,728 | {"max_depth": 30, "min_samples_leaf": 1, "min_samples_split": 2, "n_estimators": 300} | 0.8457 | 0.8537 | 0.8527 | 0.815 |
| E16-per_timeout-rf-60s | per_timeout_60s | 11,783 / 2,946 | {"max_depth": 30, "min_samples_leaf": 1, "min_samples_split": 2, "n_estimators": 300} | 0.8307 | 0.8316 | 0.8308 | 0.7873 |
| E16-per_timeout-xgb-120s | per_timeout_120s | 7,671 / 1,918 | fixed (configs/final/B.yaml) | | 0.8384 | 0.8346 | 0.7881 |
| E16-per_timeout-xgb-15s | per_timeout_15s | 14,459 / 3,615 | fixed (configs/final/B.yaml) | | 0.8924 | 0.8919 | 0.8637 |
| E16-per_timeout-xgb-30s | per_timeout_30s | 10,910 / 2,728 | fixed (configs/final/B.yaml) | | 0.8728 | 0.872 | 0.8327 |
| E16-per_timeout-xgb-60s | per_timeout_60s | 11,783 / 2,946 | fixed (configs/final/B.yaml) | | 0.8551 | 0.8544 | 0.8076 |

| training rows (timeouts mixed) | runs | test accuracy mean | std | min - max | 15s test rows | 30s test rows | 60s test rows | 120s test rows | (b) same rows, one timeout |
|---|---|---|---|---|---|---|---|---|---|
| 32,789 | 1 | 0.8796 | | | 0.8913 | 0.9038 | 0.8982 | 0.8222 | |
| 14,459 | 5 | 0.8504 | 0.0021 | 0.8470 - 0.8529 | 0.8643 | 0.8798 | 0.8655 | 0.7887 | 15s: 0.8658 |
| 11,783 | 5 | 0.8435 | 0.0028 | 0.8401 - 0.8464 | 0.8592 | 0.8742 | 0.8558 | 0.7809 | 60s: 0.8316 |
| 10,910 | 5 | 0.8396 | 0.0024 | 0.8367 - 0.8430 | 0.8559 | 0.8708 | 0.8517 | 0.7755 | 30s: 0.8537 |
| 7,671 | 5 | 0.8243 | 0.0038 | 0.8202 - 0.8302 | 0.8402 | 0.8576 | 0.8372 | 0.7583 | 120s: 0.8206 |

**Result.**

- **(a)** Their result is reproduced: 0.8796 accuracy and 0.8792 weighted F1, against their 0.8801 and 0.8797, on the same split. Their accuracy hides weaker small classes (macro F1 0.8407).
- **(b)** With the timeouts kept apart, the same method scores 0.8206-0.8658 (mean 0.8430).
- **(d)** That drop is explained by training-set size, not by the mixing. Stacking the four files gives 2.3-4.3 times as many training rows. At equal size the mixed setup scores 0.8395 on average, slightly below (b).
- **(c)** Our tuned XGBoost beats their forest at every timeout (mean 0.8647; 0.8924 at 15 s).

The Kaggle rows use the A2 column layout and an 80/20 split, so these scores are not comparable with our validation scores. **Verdict.** (a), (b), (c) and (d) supported. E16 feeds no model selection. [E16]

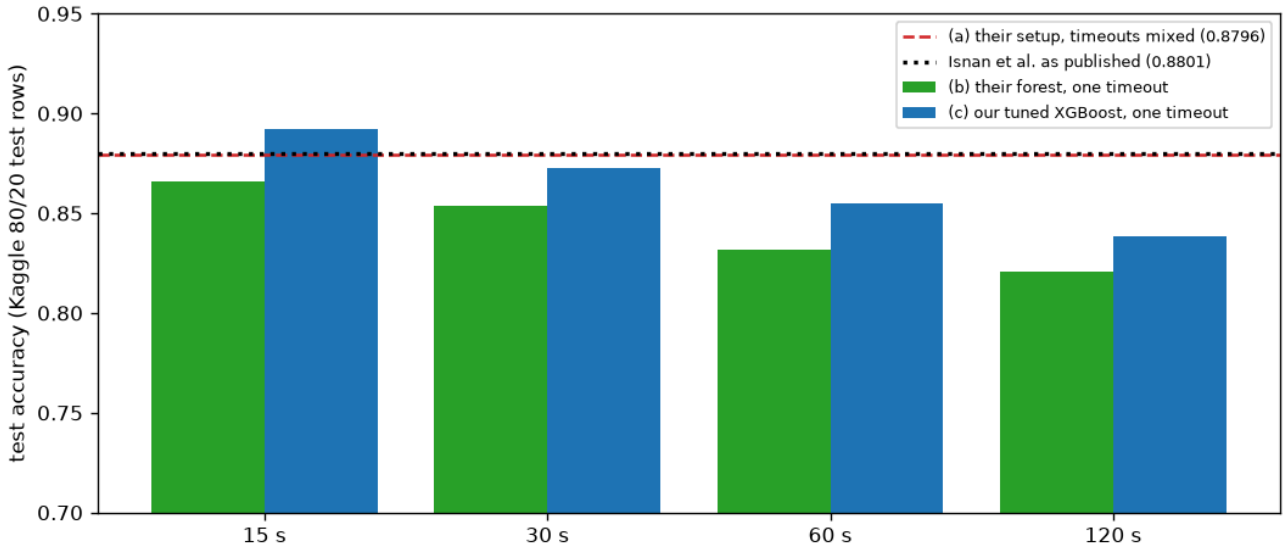



*Figure 19. E16 Isnan et al. reproduced and corrected*

## 4.19 Storage after Phase 3

Every official run saves its model (compressed, level 3). Per experiment and model type:

| kind | runs | total_MB | per_run_max_MB | model_file_max_MB |
|---|---|---|---|---|
| E04 knn_log_robust | 4 | 4.678 | 1.577 | 1.435 |
| E04 knn_log_standard | 7 | 6.54 | 1.579 | 1.437 |
| E05 rf | 7 | 102.403 | 22.6 | 22.436 |
| E06 lgbm | 4 | 4.56 | 2.168 | 2.014 |
| E06 xgb | 4 | 2.522 | 1.226 | 1.064 |
| E07 knn_log_standard | 4 | 4.78 | 1.604 | 1.437 |
| E07 lgbm | 3 | 92.646 | 62.694 | 62.51 |
| E07 rf | 1 | 3.835 | 3.835 | 3.72 |
| E07 xgb | 4 | 9.238 | 4.508 | 4.315 |
| E08 lgbm | 5 | 67.775 | 27.135 | 27.033 |
| E08 xgb | 2 | 4.724 | 4.511 | 4.348 |
| E09 lgbm | 39 | 482.749 | 27.644 | 27.549 |
| E09 xgb | 13 | 61.881 | 5.261 | 5.092 |
| E10 xgb_two_step | 1 | 3.856 | 3.856 | 3.728 |
| E13 lgbm | 75 | 474.7 | 21.265 | 21.163 |
| E13 xgb | 25 | 73.528 | 4.188 | 4.025 |
| E14 lgbm | 30 | 372.146 | 25.859 | 25.759 |
| E14 xgb | 10 | 44.771 | 4.494 | 4.336 |
| E16 rf | 26 | 453.605 | 35.049 | 34.902 |
| E16 xgb | 4 | 17.765 | 4.829 | 4.697 |
| E17 lgbm | 1 | 7.215 | 7.215 | 7.092 |
| E17 logreg | 4 | 0.515 | 0.199 | 0.004 |
| E17 rf | 3 | 68.011 | 45.962 | 45.773 |
| E17 svm_rbf | 4 | 2.944 | 1.479 | 1.306 |

| kind | runs | total_MB | per_run_max_MB | model_file_max_MB |
|---|---|---|---|---|
| E17 tree | 4 | 0.732 | 0.275 | 0.07 |

Whole store at this point: artifacts 2645.3 MiB, mlflow.db 35.8 MiB. This is far below the plan's 20 GB limit. The four headline model files are 4.31-25.12 MiB, all below GitHub's 50 MB warning size.